

Cryptography & Network Security

CHAPTER 1

Introduction to Cryptography

Neso Academy





Cryptography and Network Security

Introduction



Prerequisite

- ★ Computer Networks.
- ★ Mathematical background.

Target Audience

- ★ Undergraduate students.
- ★ Academic and Professional audiences.
- ★ Aspirants to crack competitive examinations/GATE.
- ★ Ethical hackers, Cyber security aspirants, Cyber forensics aspirants, IT/Network administrators.



Syllabus

Chapter 1: Introduction to Cryptography

Chapter 2: Mathematical Background - Abstract Algebra & Number Theory

Chapter 3: Block Ciphers

Chapter 4: Public Key Cryptography

Chapter 5: Cryptographic Hash functions and Digital Signatures

Chapter 6: Security Practices and System Security

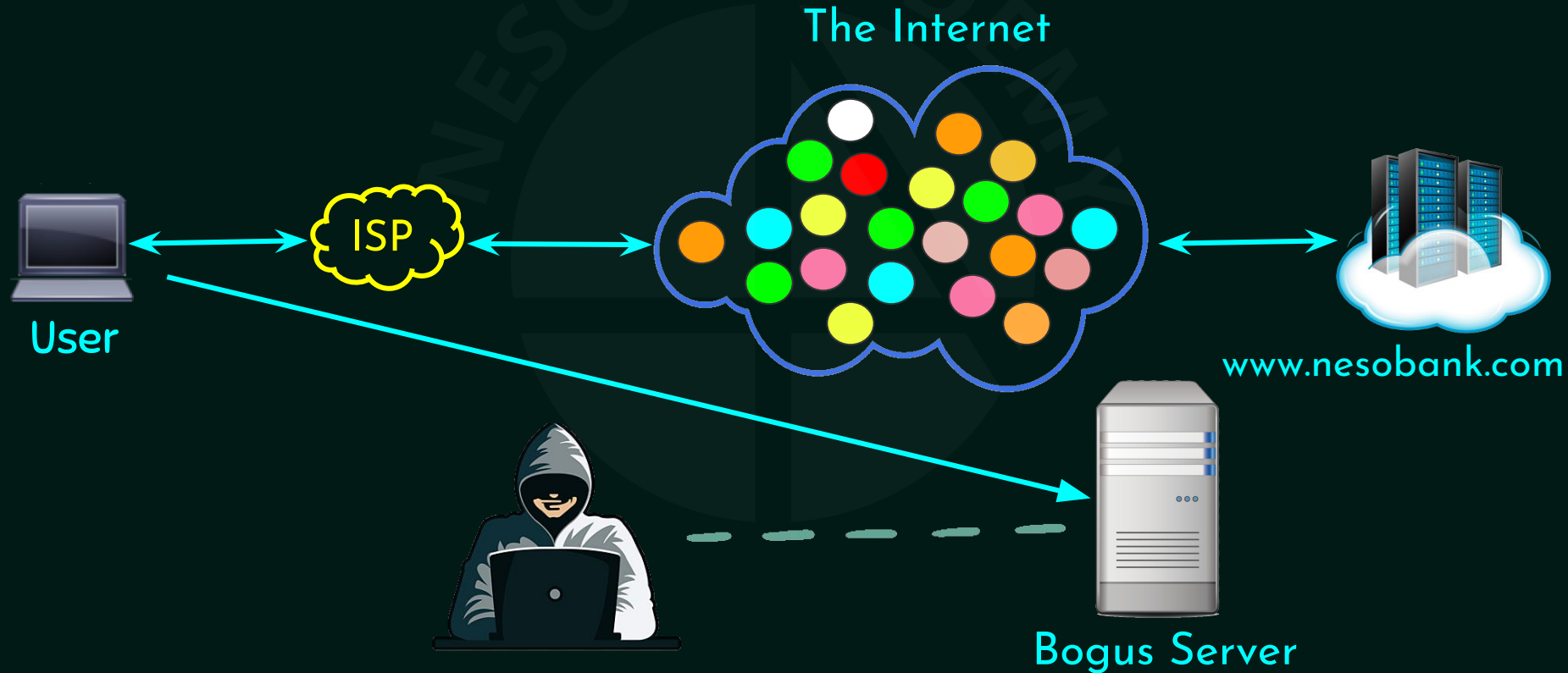
Chapter 7: Email, IP and Web Security

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand the need for network security with a real world scenario.
- ★ Know various intentions of attacker.

Need for Network Security





Cryptography and Network Security

CIA Triad

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Define computer security.
- ★ Know the key objectives of computer security.
- ★ Understand the CIA triad.
- ★ Know various levels of impact of security breach.

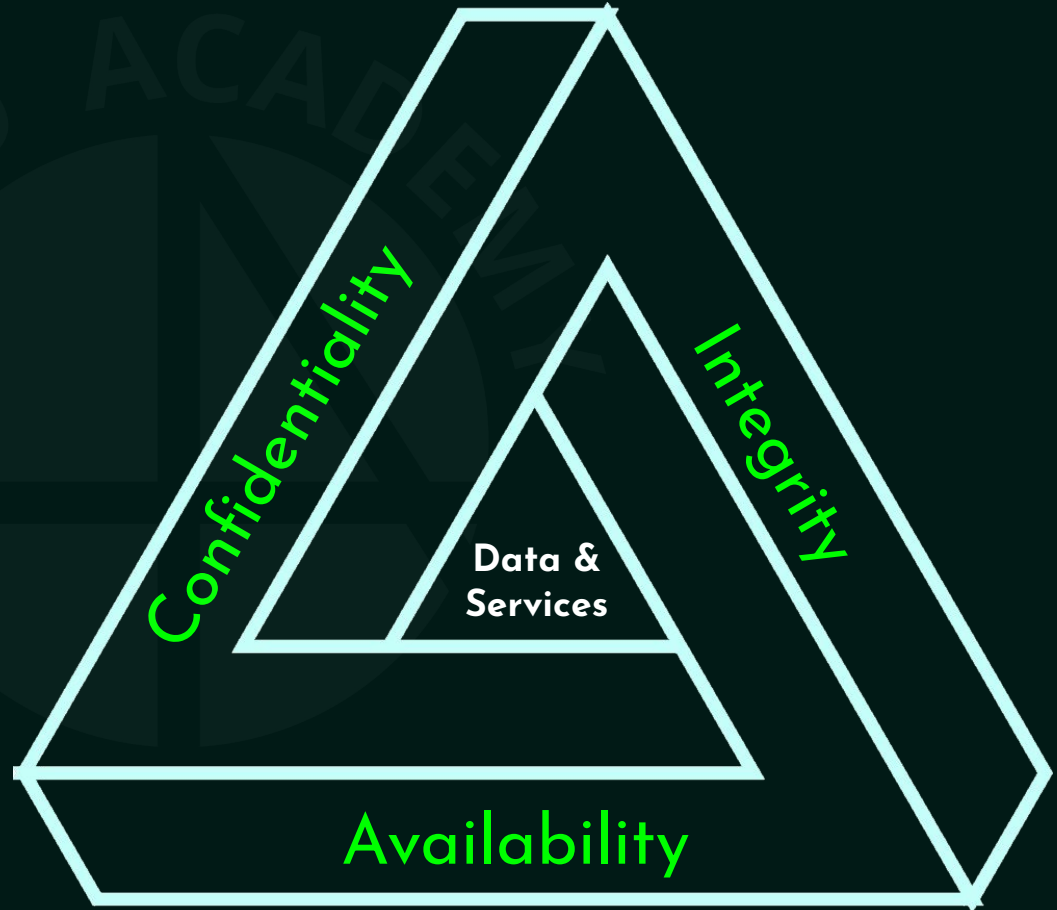
Computer Security – Definition

The protection afforded to an automated information system in order to attain the applicable objectives of preserving the integrity, availability, and confidentiality of information system resources (includes hardware, software, firmware, information/data, and telecommunications).

-
NIST

CIA Triad

- ★ Confidentiality
- ★ Integrity
- ★ Availability



Levels of impact of security breach

- ★ Low
- ★ Medium
- ★ High

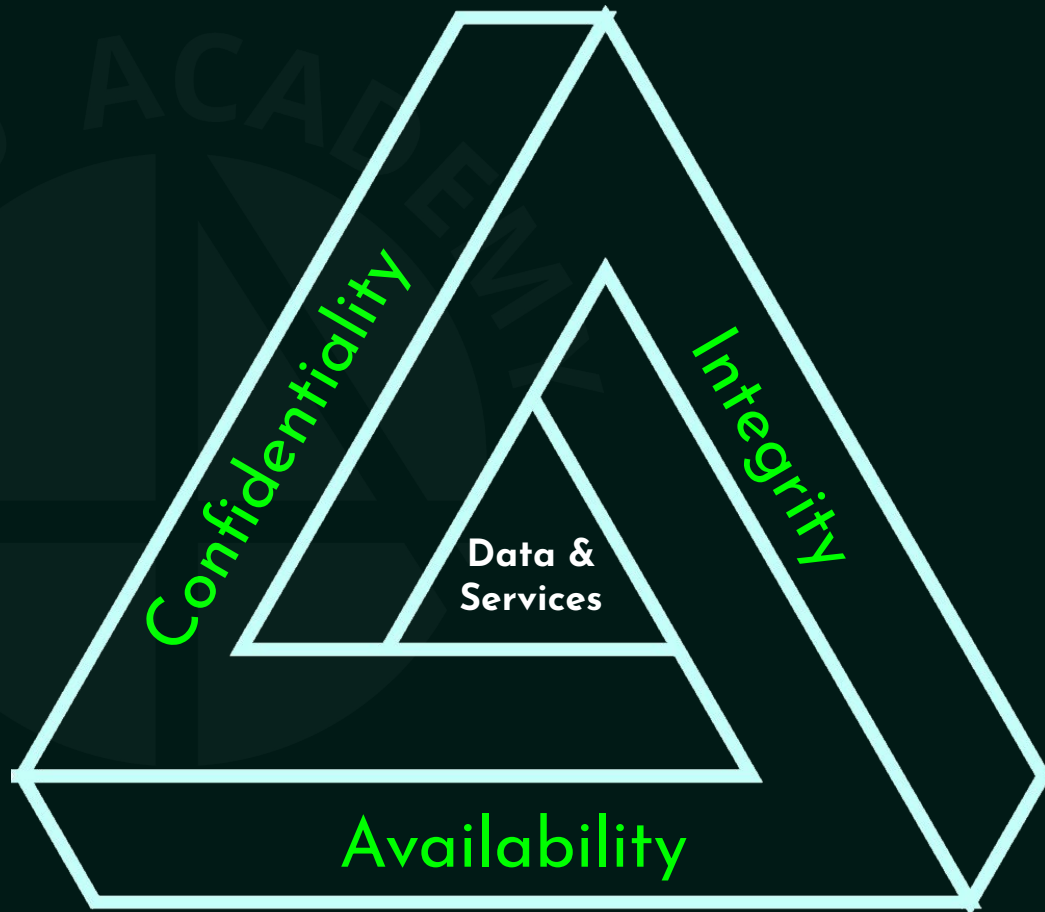


CIA Triad

- ★ Confidentiality
- ★ Integrity
- ★ Availability

Additional:

- ★ Authenticity
- ★ Accountability



CIA Triad

- ★ Confidentiality (Example: Account information)
- ★ Integrity (Example: Patient's information)
- ★ Availability (Example: Authentication service)



Cryptography and Network Security

The OSI Security Architecture

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Know the difference between threat and attack.
- ★ Understand the OSI security architecture.
- ★ Know about security attacks, security services and security mechanisms.

Analogy



Threats and Attacks (RFC 2828)

Threat: A potential for violation of security, which exists when there is a circumstance, capability, action, or event that could breach security and cause harm. That is, a threat is a possible danger that might exploit a vulnerability.

Attack: An assault on system security that derives from an intelligent threat; that is, an intelligent act that is a deliberate attempt (especially in the sense of a method or technique) to evade security services and violate the security policy of a system.

The OSI Security Architecture

- ★ Security Attack
- ★ Security Mechanism
- ★ Security Service

The OSI Security Architecture

Security Attack: Action that compromises the security.

Security Mechanism: Detect, prevent, or recover from a security attack.

Security Service: Enhances the security, counter security attacks, and provide the service.

Security Attacks

- ★ Passive attacks
- ★ Active attacks



Security Services

- ★ Authentication
- ★ Access control
- ★ Data confidentiality
- ★ Data Integrity
- ★ Nonrepudiation

Security Mechanisms

- ★ Encipherment
- ★ Digital Signature
- ★ Access Control
- ★ Data Integrity
- ★ Authentication Exchange
- ★ Traffic Padding
- ★ Routing Control
- ★ Notarization



Cryptography and Network Security

Security Attacks

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand various passive attacks.
- ★ Understand various active attacks.
- ★ Know the difference between passive and active attacks.

The OSI Security Architecture

- ★ Security Attack
- ★ Security Mechanism
- ★ Security Service

Security Attack

- ★ Action that compromises the security of an individual or an organization.

Types:

1. Passive attack
2. Active attack

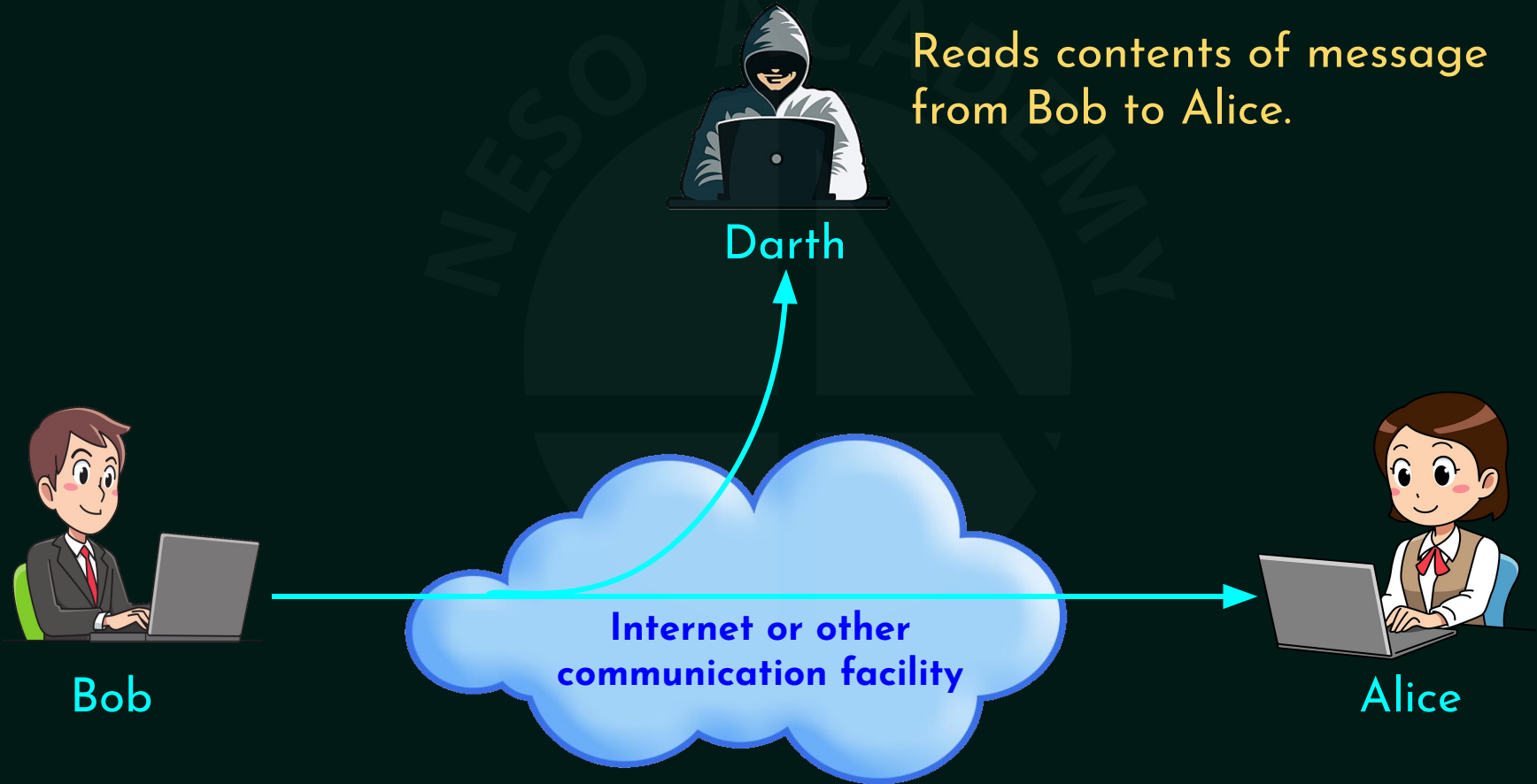
Passive Attack

- ★ Attempts to learn or make use of information from the system.
- ★ Does not affect system resources.
- ★ Eavesdropping or monitoring of transmissions.
- ★ Goal: Obtain information that is being transmitted.

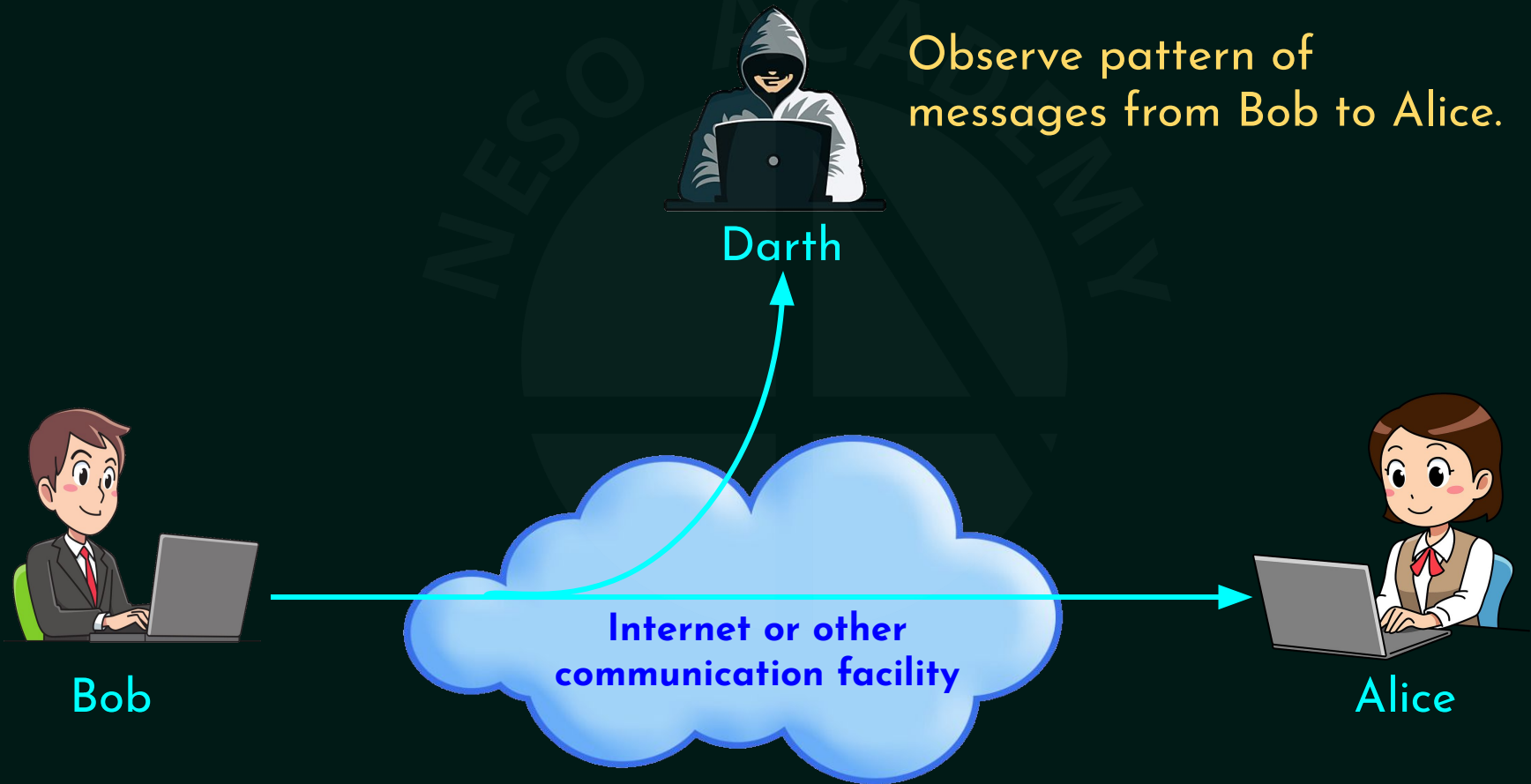
Types:

1. Release of message contents
2. Traffic Analysis

Release of message contents



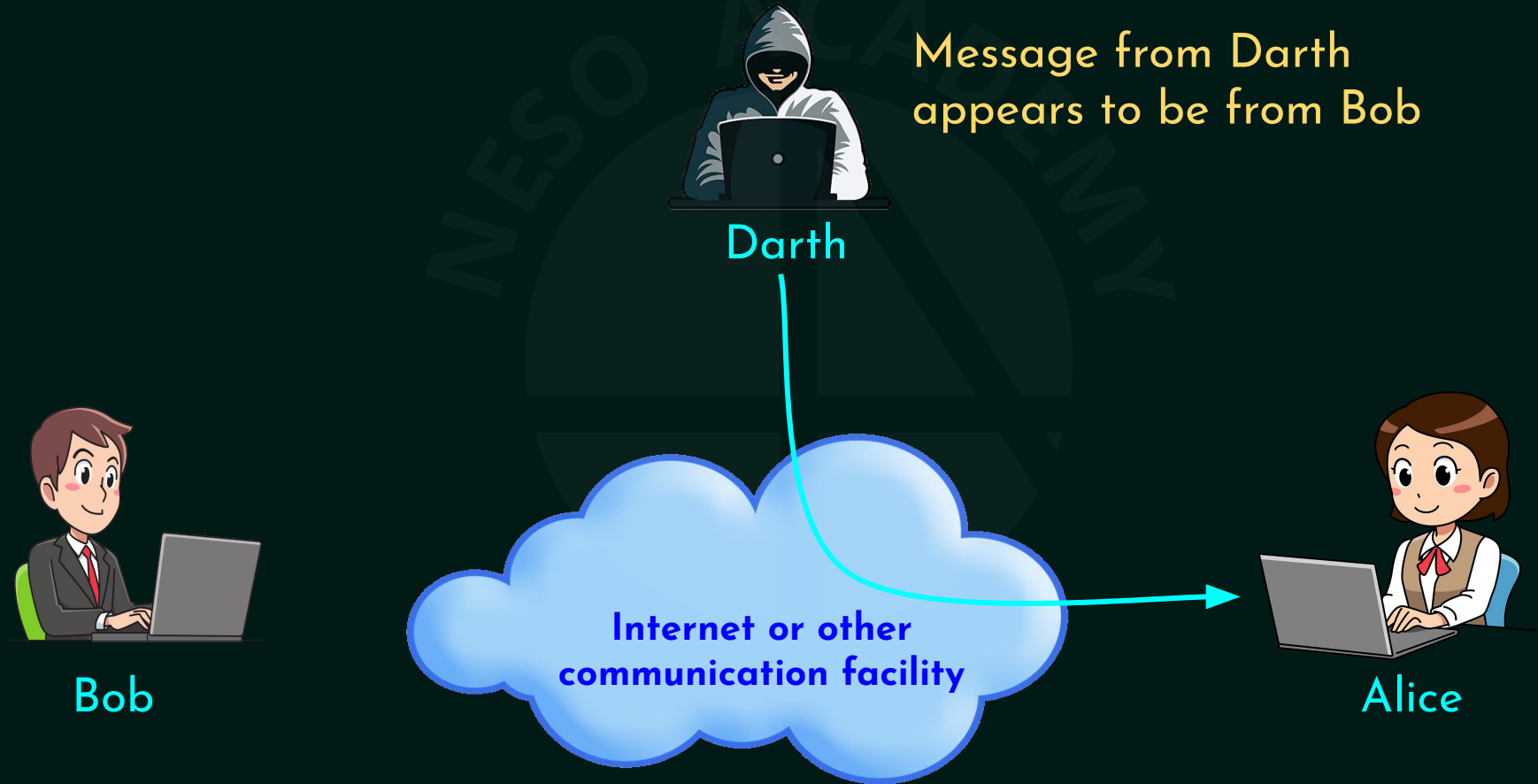
Traffic Analysis



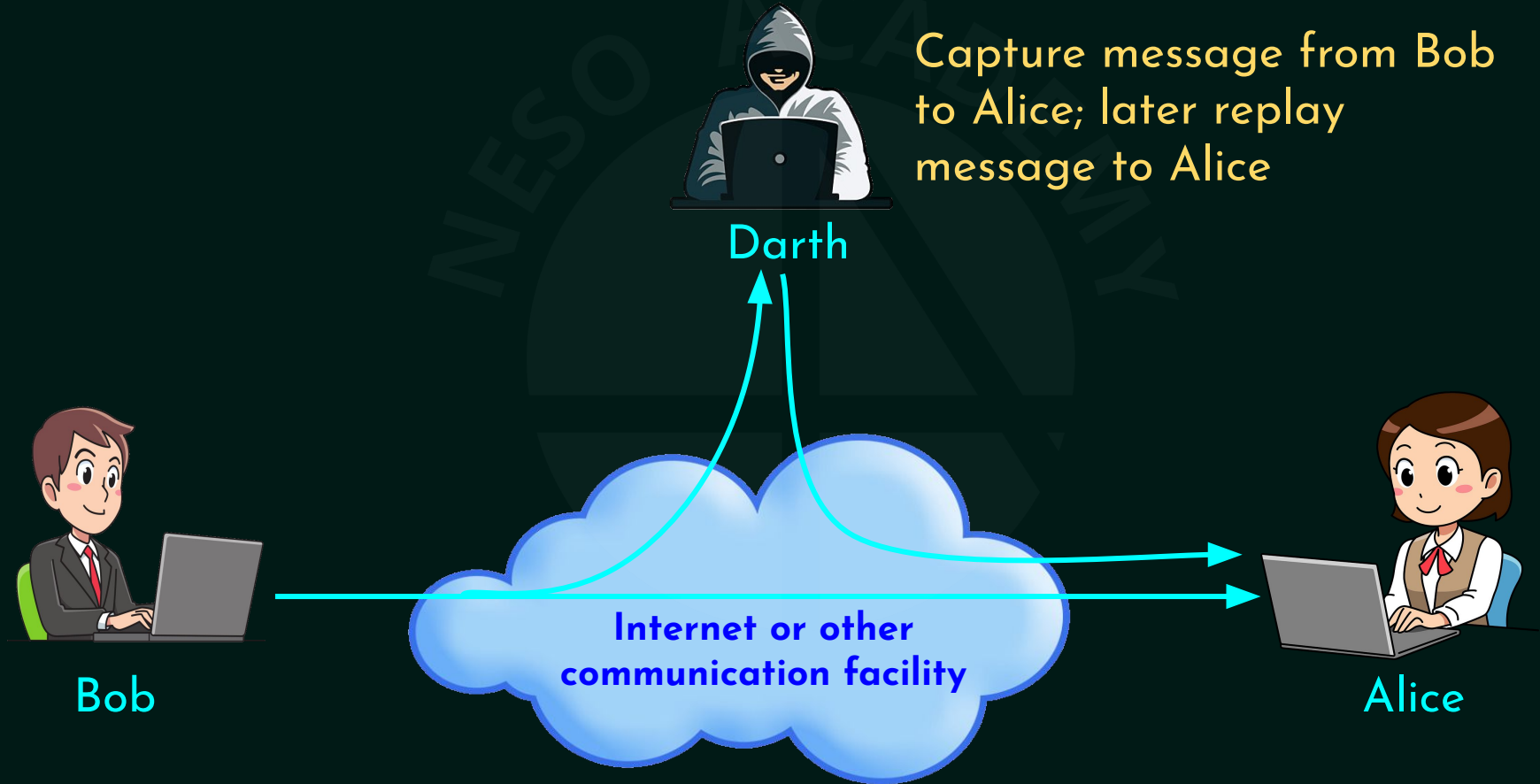
Active Attack

- ★ Active attacks involve some modification of the data stream or the creation of a false stream.
- ★ Subdivided into four categories
 1. Masquerade
 2. Replay
 3. Modification of messages
 4. Denial of Service (DoS).

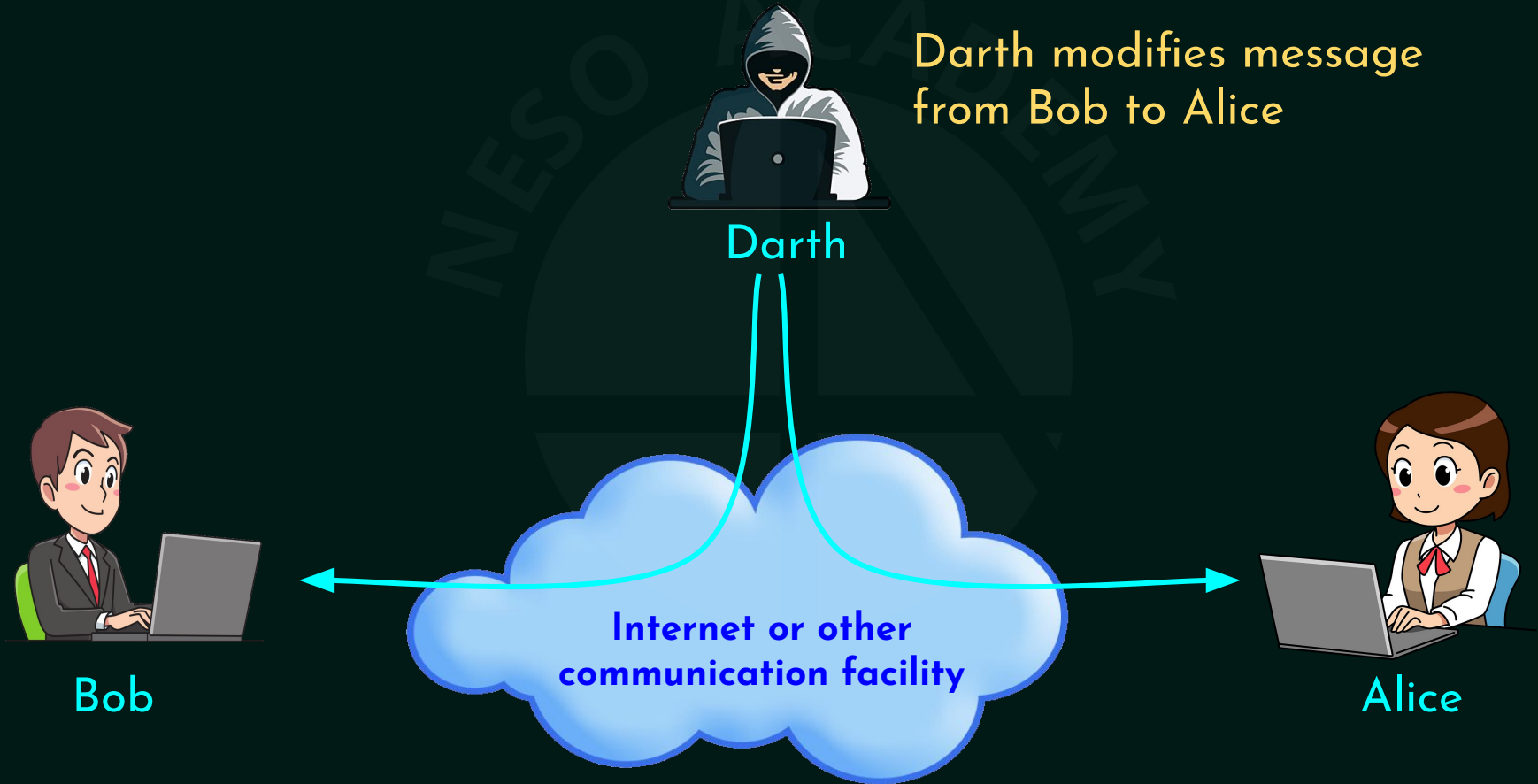
Masquerade



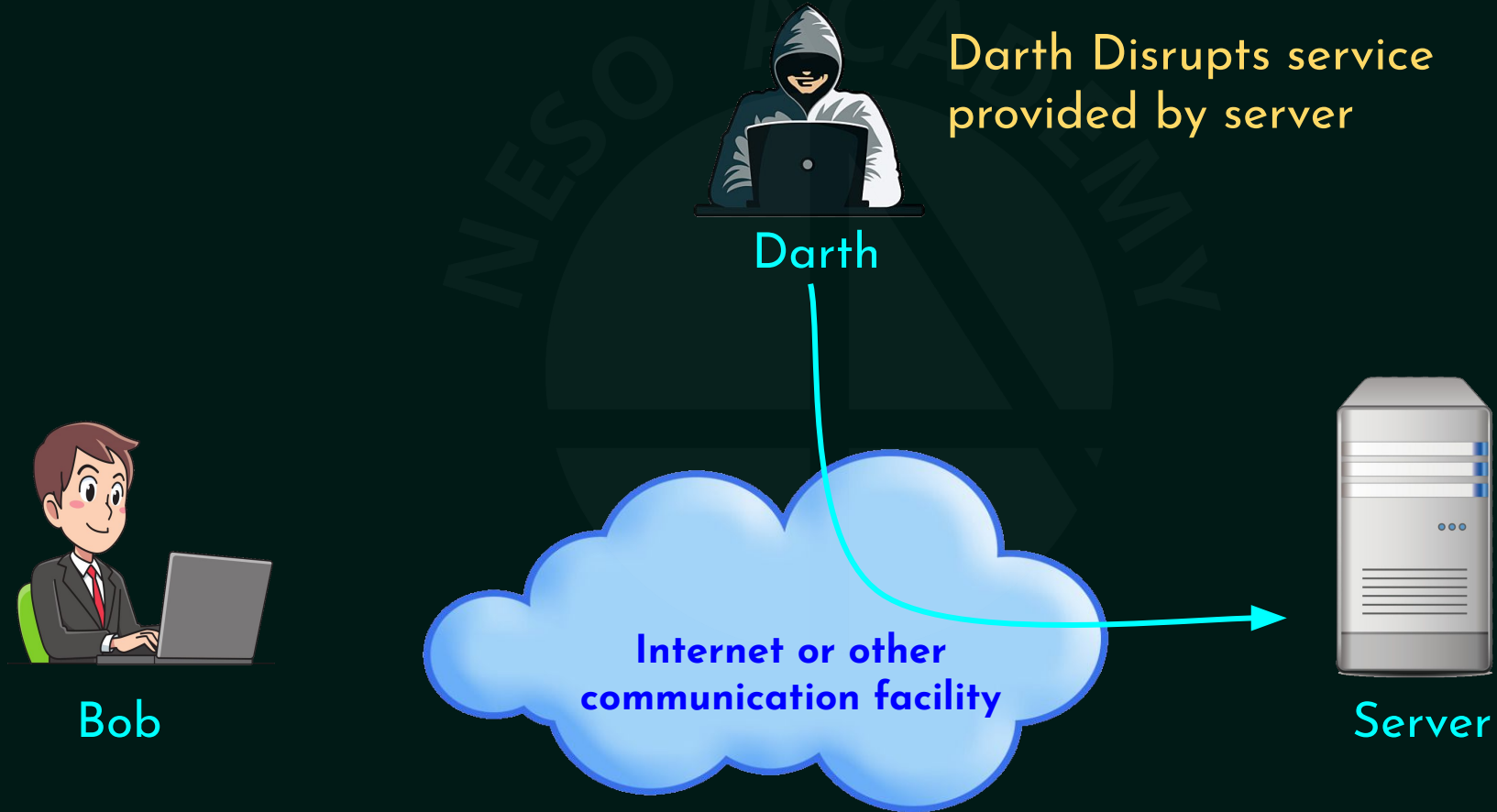
Replay



Modification of message



Denial of Service (DoS)



Passive attack Vs Active attack

Passive Attack

- ★ Hard to Detect.
- ★ Neither sender nor receiver is aware of the attack.
- ★ Encryption prevents the success of the passive attacks.
- ★ More emphasis is on prevention than detection.

Active Attack

- ★ Hard to Prevent.
- ★ Difficult to prevent - Physical, software and network vulnerabilities.
- ★ Detect and recover from any disruption or delays.
- ★ If the detection has a deterrent effect, it may also contribute to prevention.



Cryptography and Network Security

Security Services

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand various security services.

The OSI Security Architecture

- ★ Security Attack
- ★ Security Mechanism
- ★ Security Service

Security Services – Definition

The processing or communication service that is provided by a system to give a specific kind of protection to system resources; security services implement security policies and are implemented by security mechanisms.

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RFC 2828

Security Services

- ★ Authentication
 - Peer entity authentication
 - Data origin authentication
- ★ Access control
- ★ Data confidentiality
- ★ Data Integrity
- ★ Nonrepudiation

Security Services





Cryptography and Network Security

Security Mechanisms

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand various security mechanisms.

The OSI Security Architecture

- ★ Security Attack
- ★ Security Mechanism
- ★ Security Service

Specific Security Mechanisms

- ★ Encipherment
- ★ Digital Signature
- ★ Access Control
- ★ Data Integrity
- ★ Authentication Exchange
- ★ Traffic Padding
- ★ Routing Control
- ★ Notarization

Pervasive Security Mechanisms

- ★ Trusted Functionality
- ★ Security Label
- ★ Event Detection
- ★ Security Audit Trail
- ★ Security Recovery

Security Mechanisms





Cryptography and Network Security

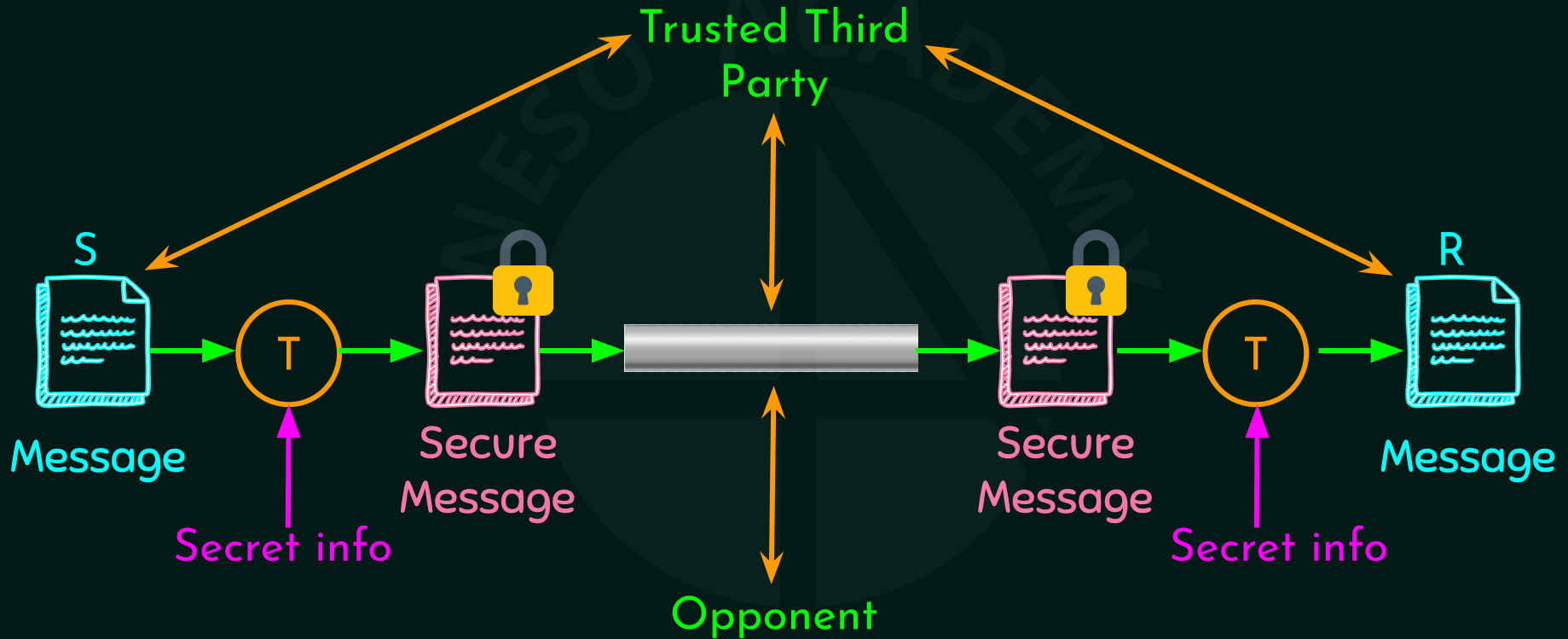
Network Security Model

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand the model for network security.
- ★ Know the four basic tasks in designing a particular security service.
- ★ Understand the network access security model.

Model for Network Security



Legends used

T - Security related transformation; S - Sender; R - Receiver

Model for Network Security

Four major tasks:

1. Design an algorithm.
2. Generate the secret information.
3. Develop methods for distribution and sharing of information.
4. Specify a protocol.

Network Access Security Model

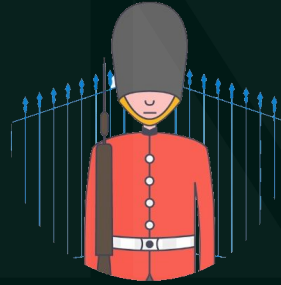


Opponent

Human (Attacker)
Software (Virus, Worm)



**Access
channel**



**Gatekeeper
function**

Information System

Computing Resources
(Processor, Memory, I/O)
Data
Process
Software

Internal Security controls



Cryptography and Network Security

Cryptography

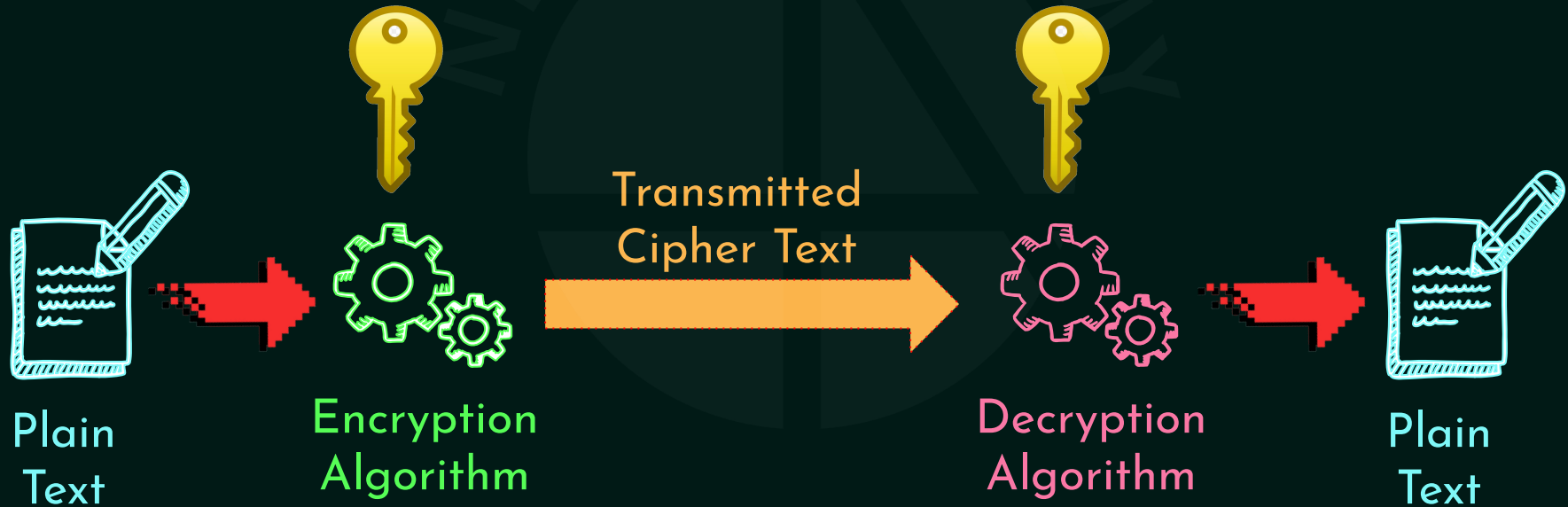
Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand the symmetric cipher model.
- ★ Understand the basics and types of cryptography.
- ★ Know the importance of encryption and decryption.
- ★ Understand unconditionally secure and computationally secure encryption schemes.

Cryptography

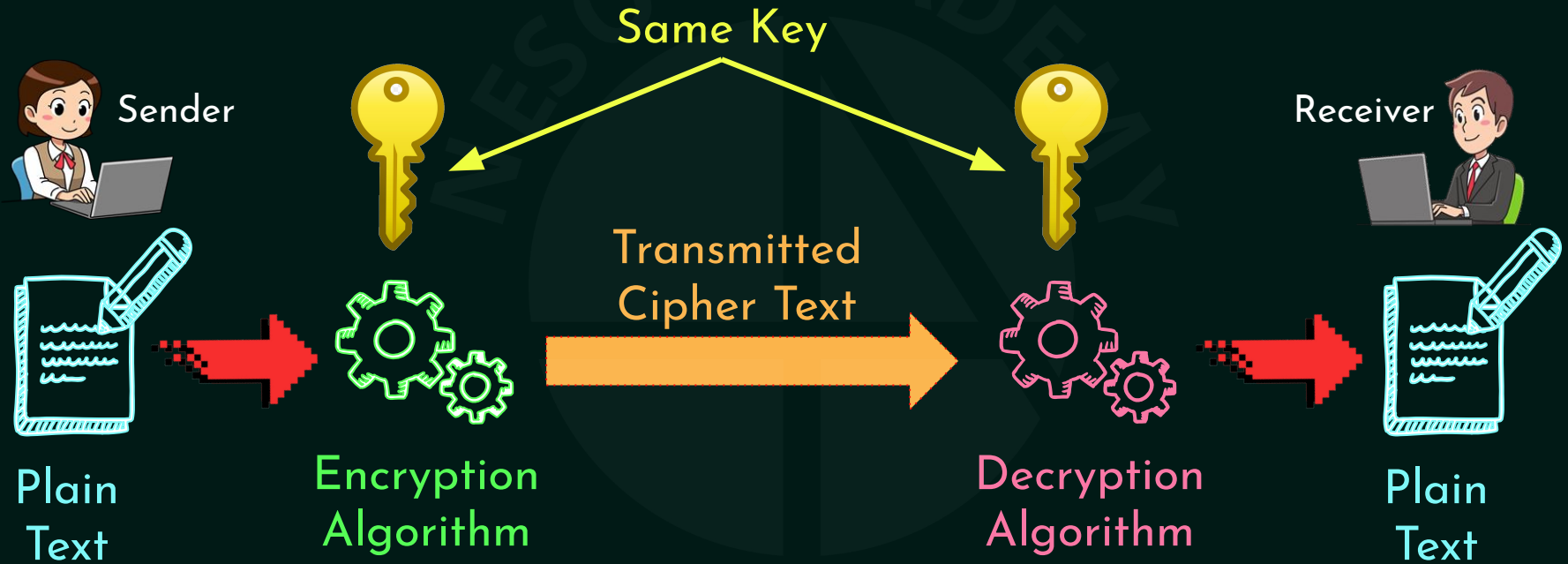
“The art or science encompassing the principles and methods of transforming an intelligible message into one that is unintelligible, and then retransforming that message back to its original form.”



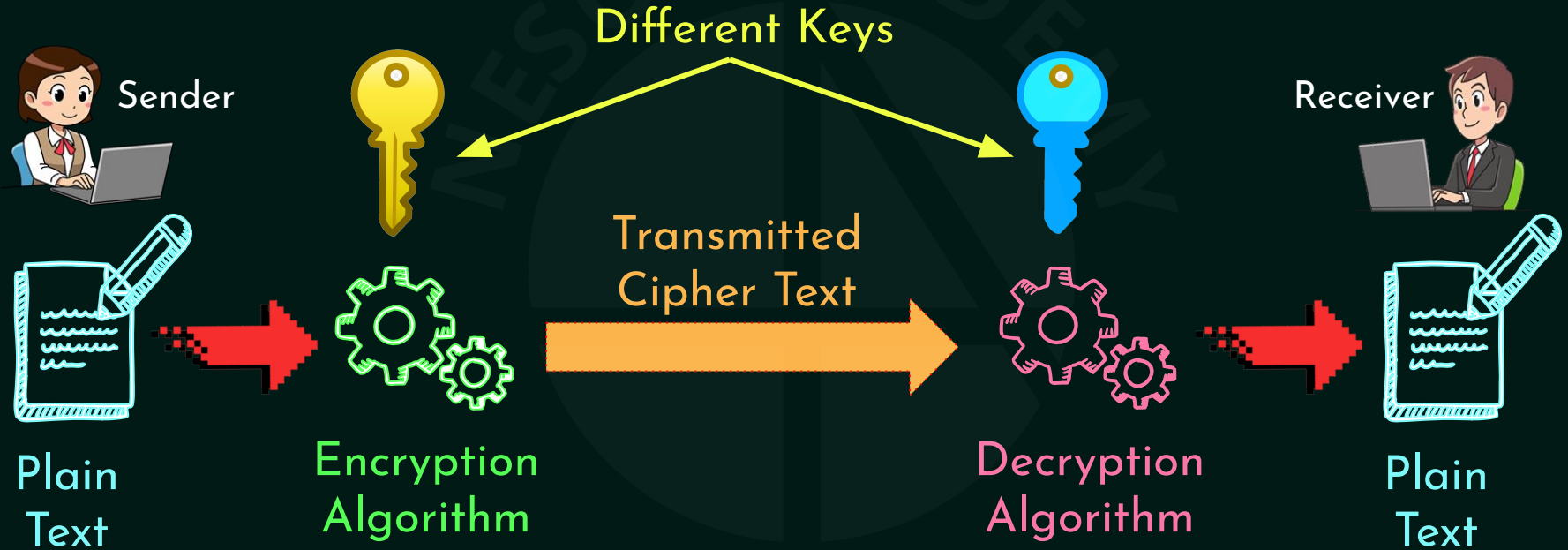
Types of Cryptography

- ★ Symmetric Cryptography (Private Key Cryptography)
- ★ Asymmetric Cryptography (Public Key Cryptography)

Symmetric Cryptography

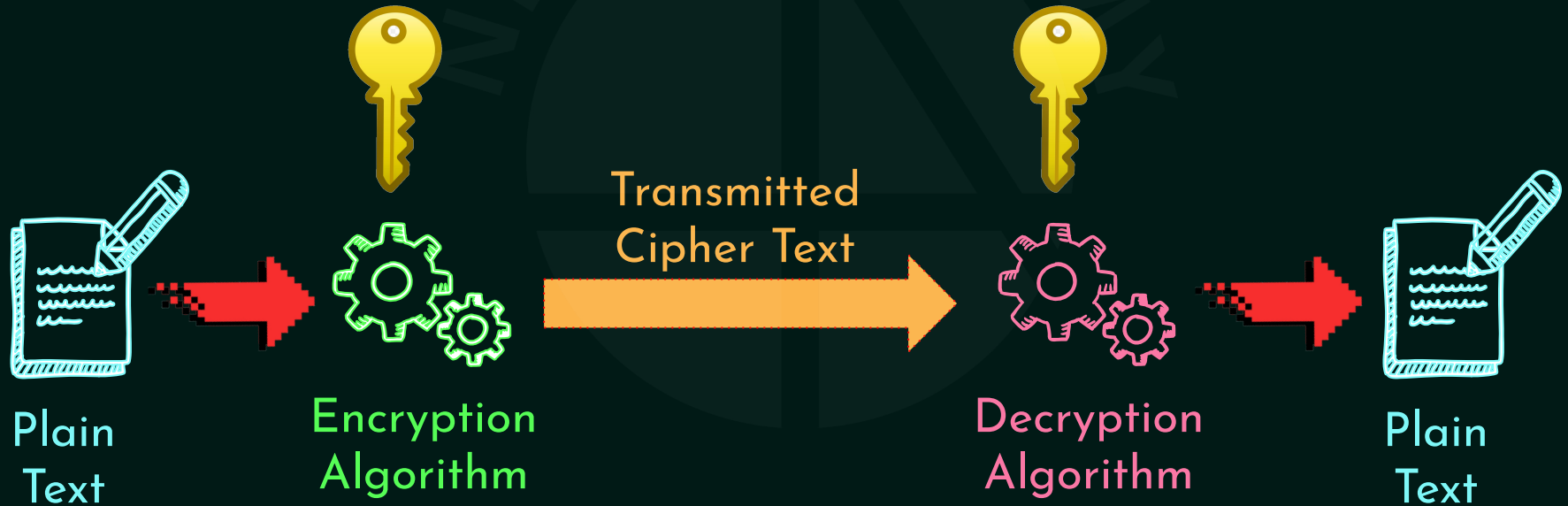


Asymmetric Cryptography



Encryption schemes

- ★ Unconditionally secure
- ★ Computationally secure





Cryptography and Network Security

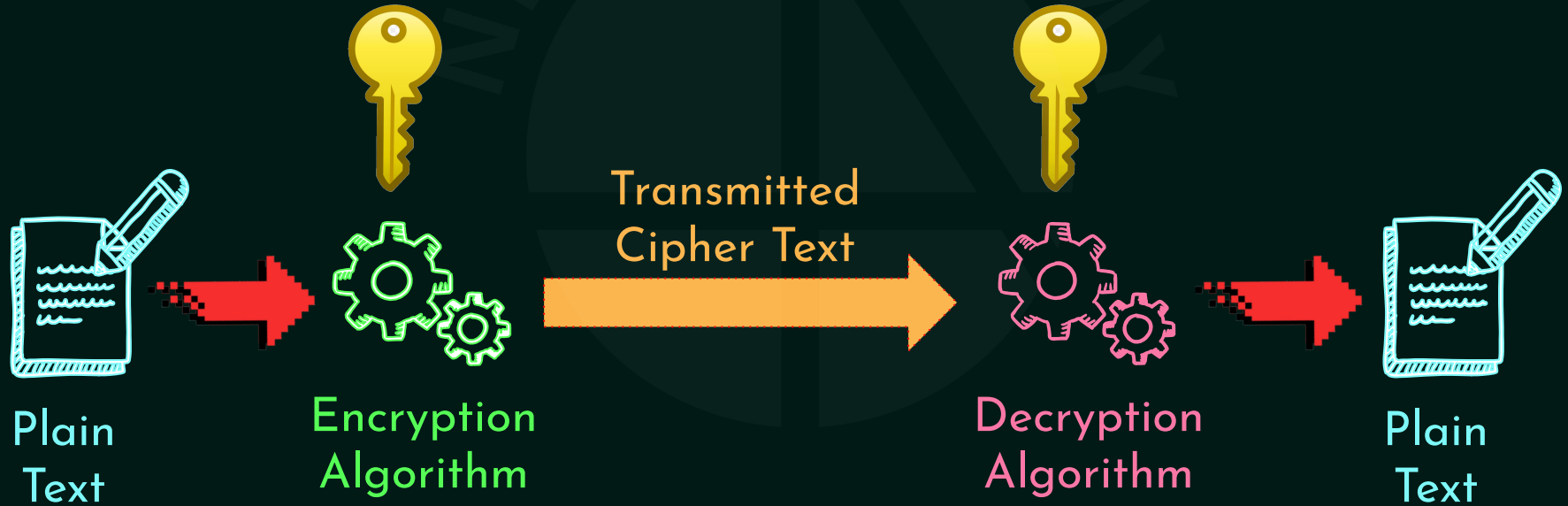
Cryptography – Key Terms

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand various key terms related to cryptography.

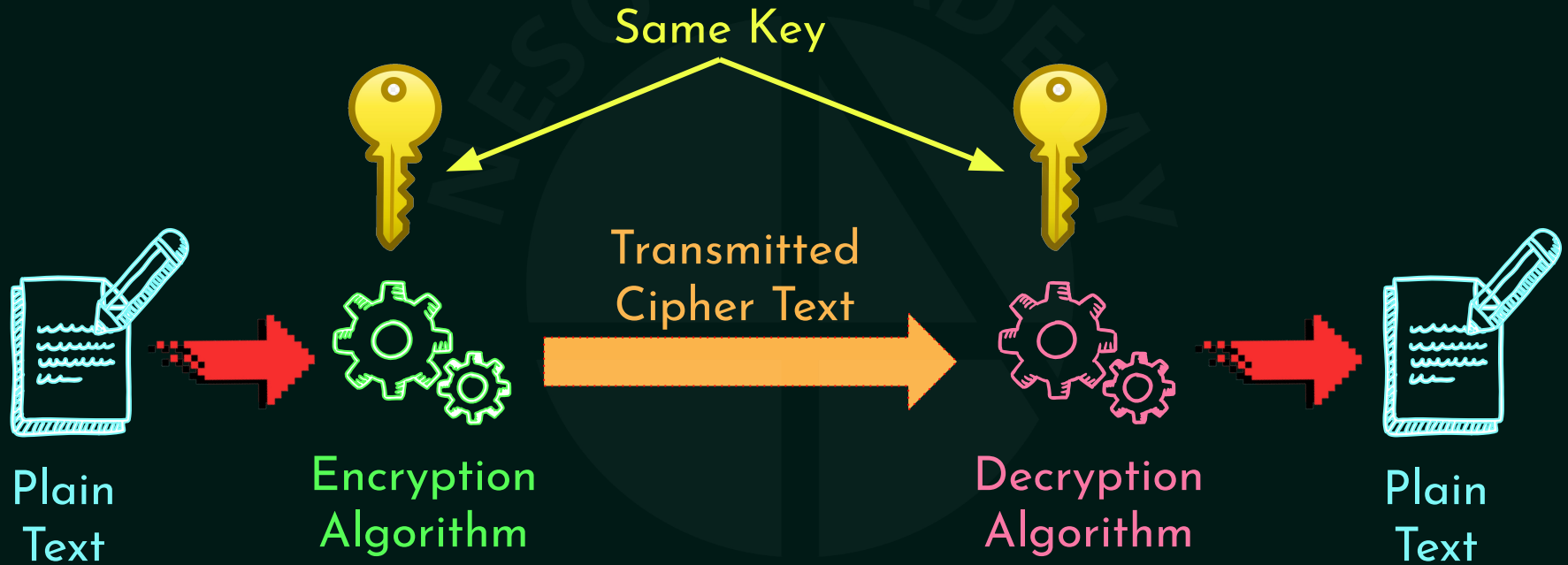
Cryptography



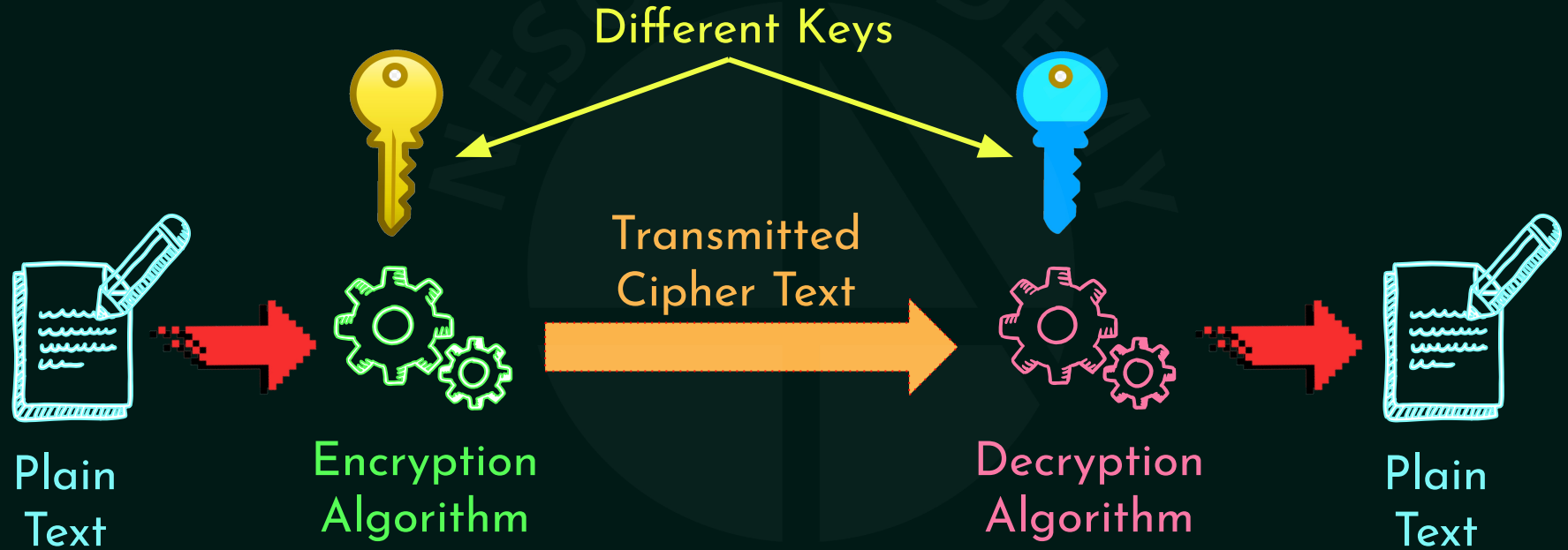
Types of Cryptography

- ★ Symmetric Cryptography (Private Key Cryptography)
- ★ Asymmetric Cryptography (Public Key Cryptography)

Symmetric Cryptography



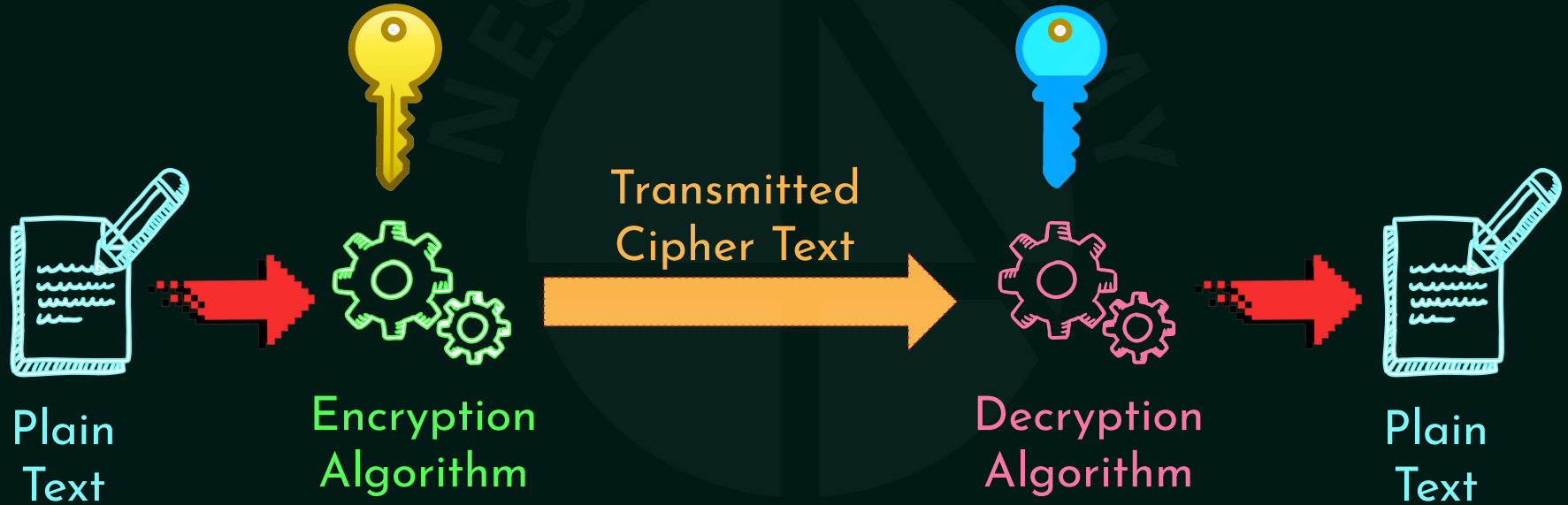
Asymmetric Cryptography



Key Terms

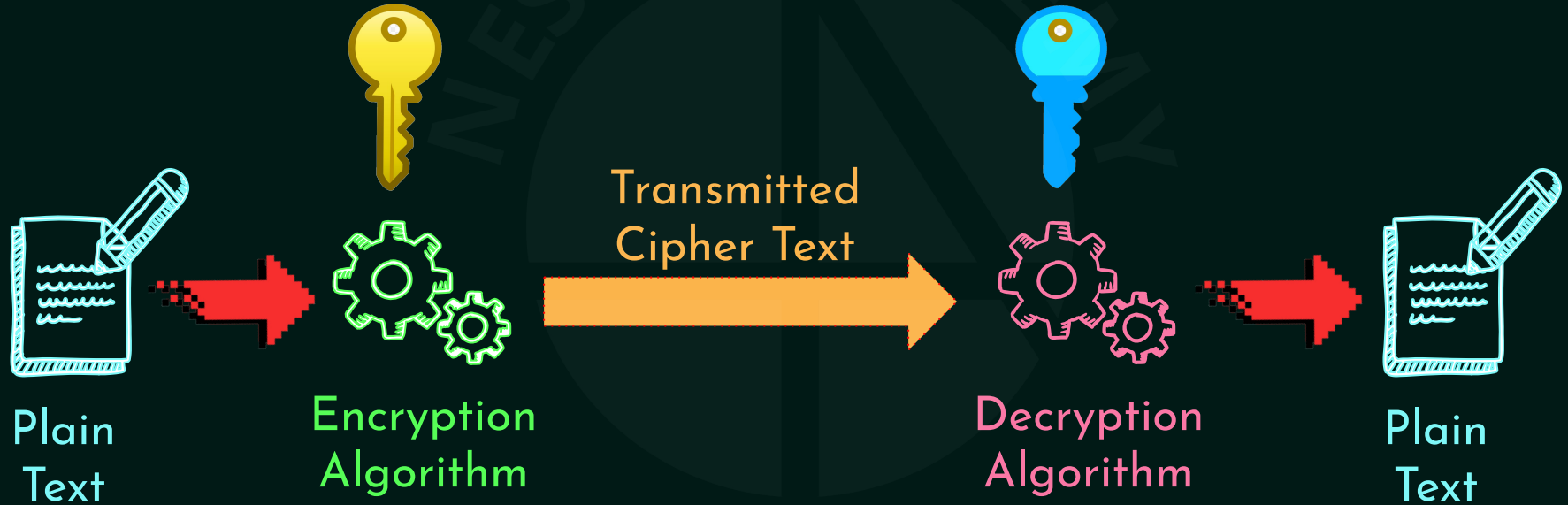
- ★ Cryptography
- ★ Plaintext
- ★ Ciphertext
- ★ Cipher (Encryption Algorithm)
- ★ Key
- ★ Cryptanalysis (code breaking)
- ★ Cryptology = Cryptography + Cryptanalysis

Key Terms



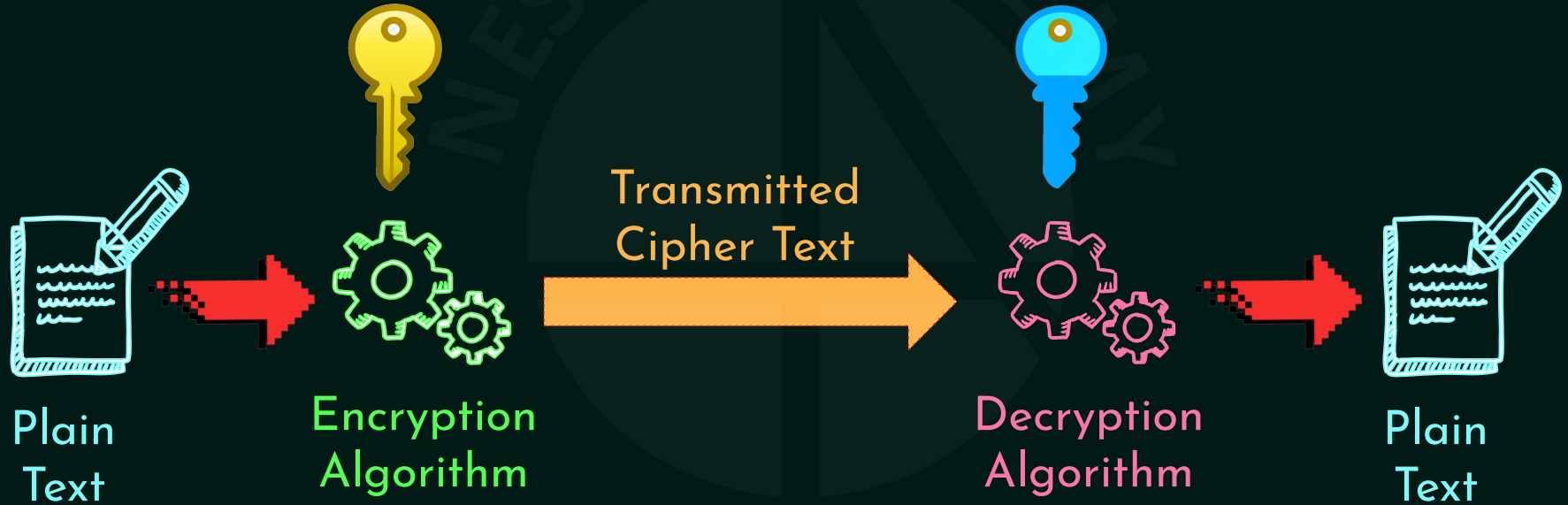
Key Term: **Cryptography**

Key Terms



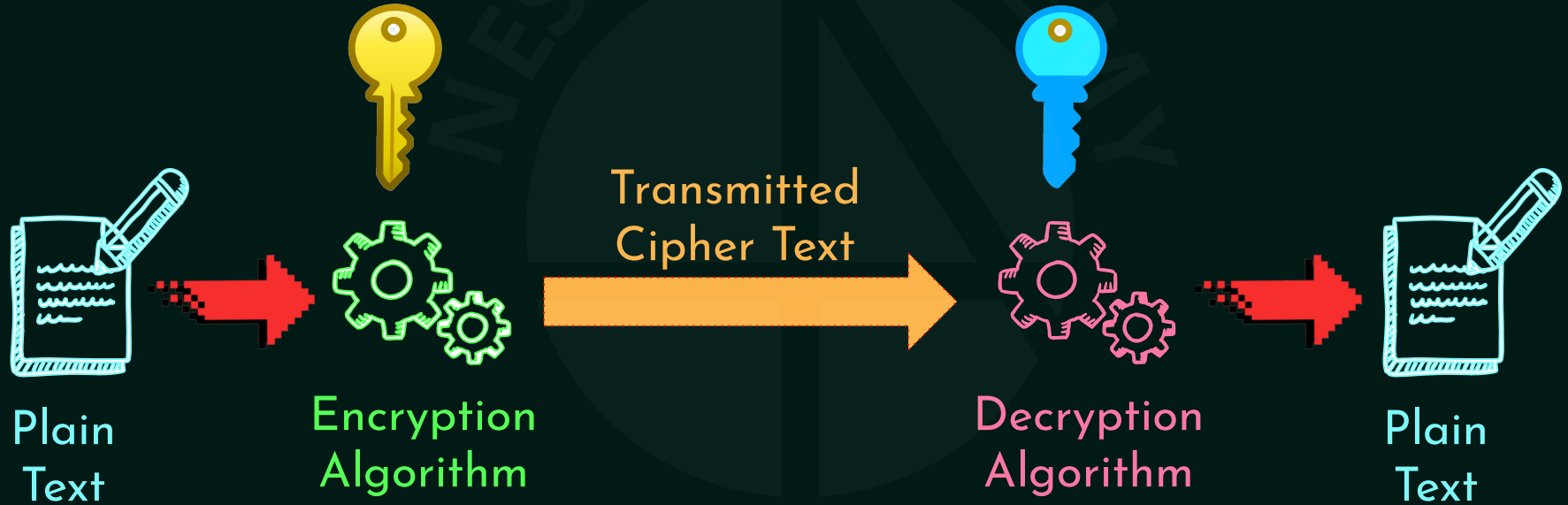
Key Term: Plaintext

Key Terms



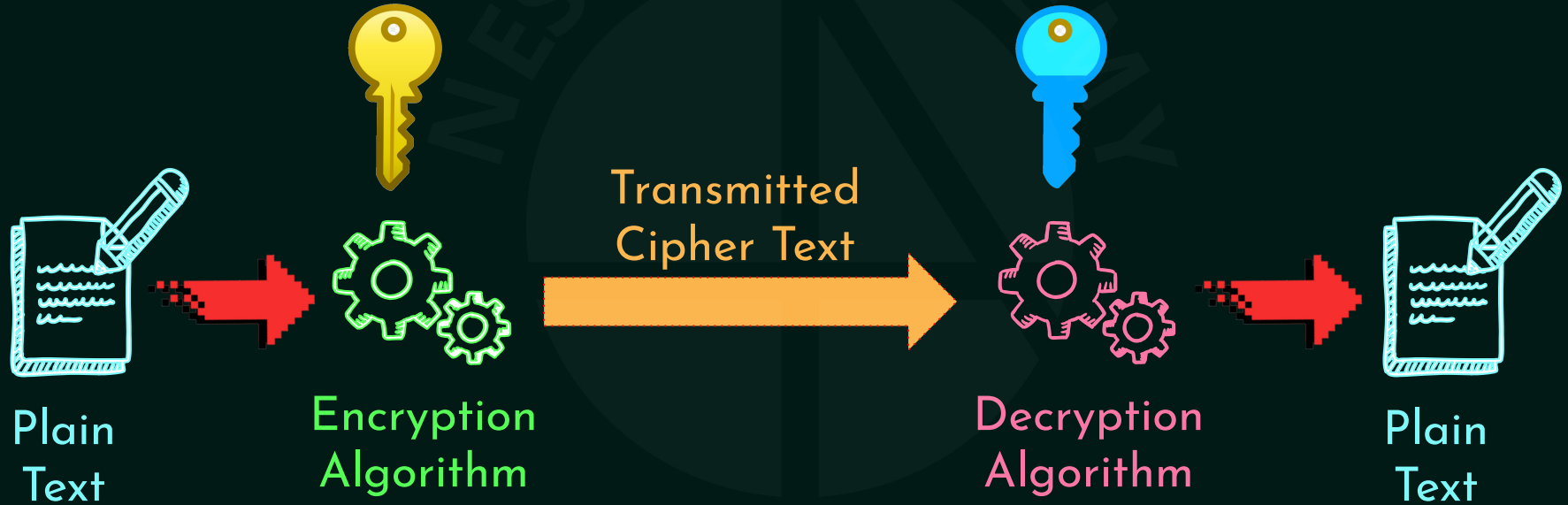
Key Term: Ciphertext

Key Terms



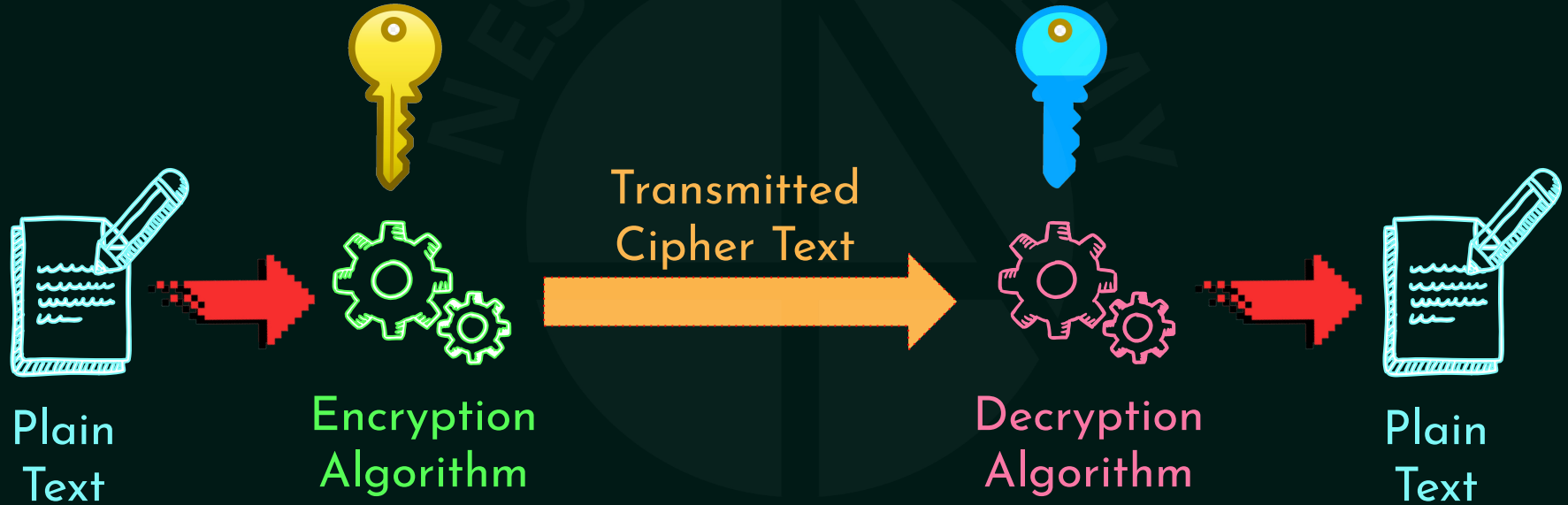
Key Term: **Cipher**

Key Terms



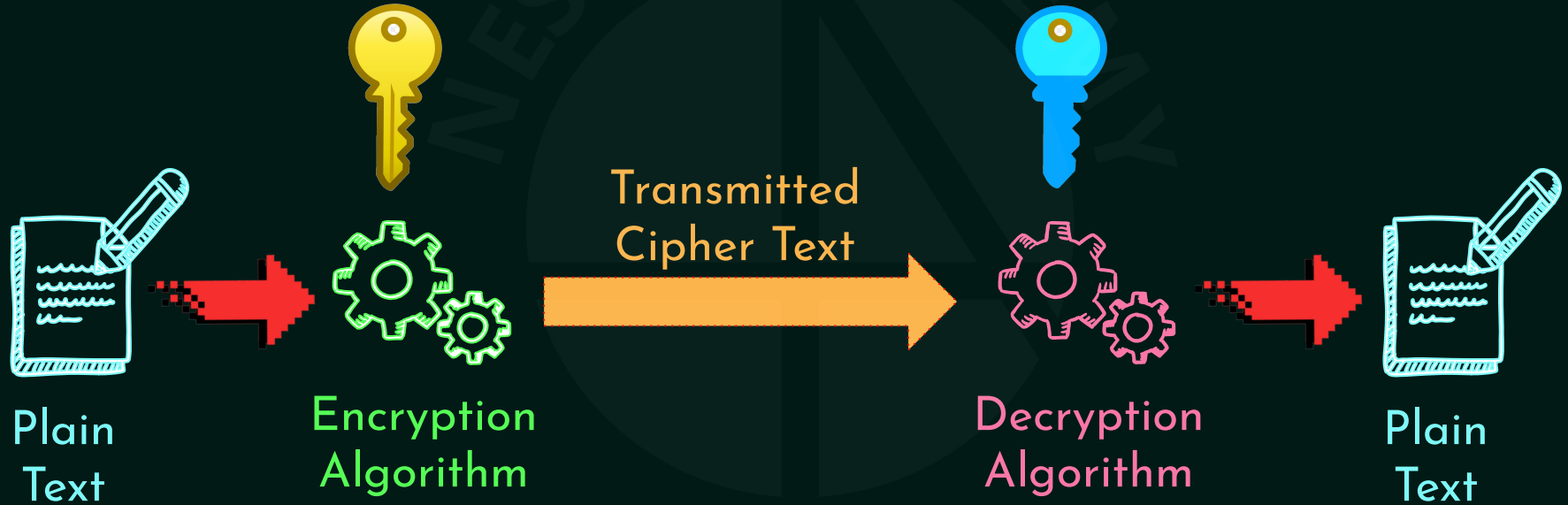
Key Term: Key

Key Terms



Key Term: Cryptanalysis

Key Terms



Key Term: Cryptology



Cryptography and Network Security

Cryptanalysis

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Know the general approaches to attacking a conventional encryption schemes.
- ★ Know about brute-force attack.
- ★ Understand various cryptanalytic attacks.

Ciphertext – Plaintext

Ciphertext

2D570755676DFF11E71B6C8511EFE7A7D3B02A3CEE63165050AB5F
4C4D19A4AAB07656A636654C6F39A4AC0FEA2035CCDD7181C0E
BB482A6EBDAEF2AEB35CB5C325CBF0738AEC27D77BEC3938C5
90CE77F62CBDCC3EA3D03E06A386BD70BC99A843DD6B7B9753
635C919FA17FC40A3C3DCBD13633D2D56A1A073EA0E73E60C60

Plaintext

Hello World

Algorithm

RSA Algorithm

General approaches

1. Cryptanalysis
2. Brute-force attack

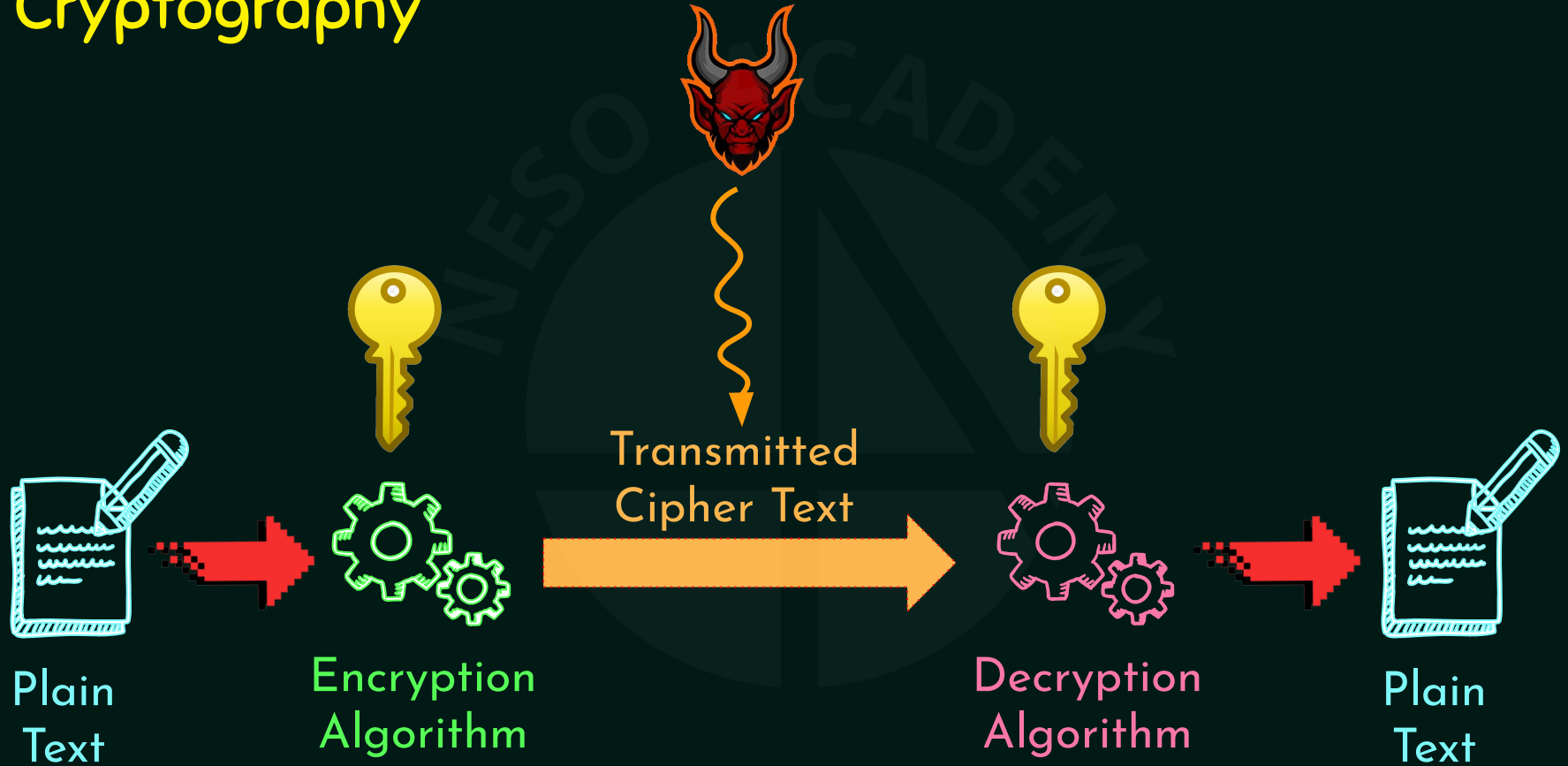


Cryptanalysis

- ★ Cryptanalytic attacks - Based on info know to the cryptanalyst.
- ★ Most difficult : Ciphertext only (Not even encryption algorithm)
- ★ Types of cryptanalytic attacks:
 1. Ciphertext Only
 2. Known Plaintext
 3. Chosen Plaintext
 4. Chosen Ciphertext
 5. Chosen Text

Type of Attack	Known to cryptanalyst
Ciphertext Only	★ Encryption Algorithm ★ Ciphertext
Known Plaintext	★ Encryption Algorithm ★ Ciphertext ★ One or more PT-CT pairs formed with secret key
Chosen Plaintext	★ Encryption Algorithm ★ Ciphertext ★ PT message chosen by cryptanalyst, together with its CT generated with the secret key
Chosen Ciphertext	★ Encryption Algorithm ★ Ciphertext ★ CT chosen by cryptanalyst, together with its corresponding decrypted PT generated with the secret key
Chosen Text	★ Chosen Plaintext and Chosen Ciphertext

Cryptography





Cryptography and Network Security

Brute Force Attack

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Know the general approaches to attacking a conventional encryption schemes.
- ★ Know about brute-force attack.
- ★ Know some software tools that can perform brute-force attack.
- ★ Understand the importance of having for CAPTCHAs.

General approaches

1. Cryptanalysis

2. Brute-force attack

Brute-force attack

- ★ Trying every possible key.
- ★ Until an intelligible translation of the ciphertext into plaintext is obtained.
- ★ Guessing.
- ★ Exhaustive key search.
- ★ Software Tools that can perform brute-force attack.

Aircrack-ng	DaveGrohl	John the ripper
Cain and Abel	Hashcat	Rainbowcrack
Crack	Hydra	Ophcrack

CAPTCHA



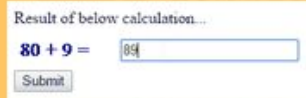
Text-based Captcha



ReCAPTCHA



3D Captcha



Mathematical Captcha



Image-based Captcha

A contrived acronym for "Completely Automated Public Turing test to tell Computers and Humans Apart"

Brute-force attack

```
[80][http-get-form] host: 192.168.100.155 login: admin password: password
[80][http-get-form] host: 192.168.100.155 login: admin password: p@ssword
[80][http-get-form] host: 192.168.100.155 login: admin password: 12345
[80][http-get-form] host: 192.168.100.155 login: admin password: 1234567890
[80][http-get-form] host: 192.168.100.155 login: admin password: Password
[80][http-get-form] host: 192.168.100.155 login: admin password: 123456
[80][http-get-form] host: 192.168.100.155 login: admin password: 1234567
[80][http-get-form] host: 192.168.100.155 login: admin password: 12345678
[80][http-get-form] host: 192.168.100.155 login: admin password: 1q2w3e4r
[80][http-get-form] host: 192.168.100.155 login: admin password: 123
[80][http-get-form] host: 192.168.100.155 login: admin password: 1
[80][http-get-form] host: 192.168.100.155 login: admin password: 12
1 of 1 target successfully completed, 12 valid passwords found
Hydra (http://www.thc.org/thc-hydra) finished at 2017-07-27 15:28:24
```

Brute-force attack

```
[*] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\DefaultAccount :stealthlagent',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\DefaultAccount :Q4)sJu\Y8qz*A3?d',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\DefaultAccount :$superP@ssword',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\WDAGUtilityAccount :stealthlagent',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\WDAGUtilityAccount :Q4)sJu\Y8qz*A3?d',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\WDAGUtilityAccount :$superP@ssword',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\None :stealthlagent',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\None :Q4)sJu\Y8qz*A3?d',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\None :$superP@ssword',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\Hazard :stealthlagent',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\Hazard :Q4)sJu\Y8qz*A3?d',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\Hazard :$superP@ssword',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\support :stealthlagent',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\support :Q4)sJu\Y8qz*A3?d',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\support :$superP@ssword',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\Chase:stealthlagent',
[+] 10.10.10.149:445 - 10.10.10.149:445 - Success: '.\Chase:Q4)sJu\Y8qz*A3?d'
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\Jason:stealthlagent',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\Jason:Q4)sJu\Y8qz*A3?d',
[-] 10.10.10.149:445 - 10.10.10.149:445 - Failed: '.\Jason:$superP@ssword',
[*] 10.10.10.149:445 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf5 auxiliary(scanner/port/port_finger) >
```

Brute-force attack

```
$ /usr/sbin/john --wordlist=/usr/share/wordlists/rockyou.txt ~/passwords.txt
Warning: detected hash type "sha512crypt", but the string is also recognized as "crypt"
Use the "--format=crypt" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 2 password hashes with 2 different salts (sha512crypt, crypt(3) $6$ [SHA512 128/128 XOP 2x])
Press 'q' or Ctrl-C to abort, almost any other key for status
password (root)
lg 0:00:00:15 0.10% (ETA: 00:14:39) 0.06410g/s 1050p/s 1054c/s 1054C/s crystal..spiderpig
lg 0:00:00:16 0.10% (ETA: 00:15:41) 0.06020g/s 1048p/s 1051c/s 1051C/s paramedic..eminem12
lg 0:00:00:34 0.21% (ETA: 00:20:41) 0.02886g/s 1053p/s 1054c/s 1054C/s 090506..zuniga
```



Cryptography and Network Security

Classical Encryption Techniques

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Know the classical encryption techniques.
- ★ Understand the difference between substitution and transposition techniques.

Classical Encryption Techniques

1. Substitution Technique
2. Transposition Technique

Substitution Technique

★ Letters are replaced by other letters or symbols.

Example:

a	b	c	d	e	f	g	h	i	j	k	l	m
n	o	p	q	r	s	t	u	v	w	x	y	z

a → M

b → X

x → Z

g → A

Plaintext : bag

Ciphertext : XMA

Transposition Technique

- ★ Applying some sort of permutation on the plaintext letters.
- ★ Plaintext: NESO
- ★ Ciphertext: ESON, SONE, ONES, ENOS

Classical Encryption Technique

Substitution	Transposition
❖ Caesar Cipher ❖ Monoalphabetic Cipher ❖ Playfair Cipher ❖ Hill Cipher ❖ Polyalphabetic Cipher ❖ One-Time Pad	❖ Rail Fence ❖ Row Column Transposition



Cryptography and Network Security

Caesar Cipher (Part 1)

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand Caesar cipher with examples.

Classical Encryption Techniques

1. Substitution Technique
2. Transposition Technique

Classical Encryption Technique

Substitution

- ❖ Caesar Cipher ←
- ❖ Monoalphabetic Substitution
- ❖ Playfair Cipher
- ❖ Hill Cipher
- ❖ Polyalphabetic Ciphers
- ❖ One-Time Pad

Transposition

- ❖ Rail Fence
- ❖ Row Column Transposition

Caesar Cipher

- ★ Letters are replaced by other letters or symbols.
- ★ The earlier known and simplest method used by Julius Caesar.
- ★ Replacing each letter of the alphabet with the letter standing three places further down the alphabet.

Example:

O	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher

Algorithm:

For each plaintext letter 'p', substitute the ciphertext letter 'C':

$$C = E(p, k) \bmod 26 = (p + k) \bmod 26$$

$$p = D(C, k) \bmod 26 = (C - k) \bmod 26$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q										

$$C = (p + k) \bmod 26$$

$$C = (13 + 3) \bmod 26$$

$$C = 16 \bmod 26$$

$$C = 16$$

$$C = Q$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H									

$$C = (p + k) \bmod 26$$

$$C = (4 + 3) \bmod 26$$

$$C = 7 \bmod 26$$

$$C = 7$$

$$C = H$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V								

$$C = (p + k) \bmod 26$$

$$C = (18 + 3) \bmod 26$$

$$C = 21 \bmod 26$$

$$C = 21$$

$$C = V$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z



Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R							

$$C = (p + k) \bmod 26$$

$$C = (14 + 3) \bmod 26$$

$$C = 17 \bmod 26$$

$$C = 17$$

$$C = R$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R	D						

$$C = (p + k) \bmod 26$$

$$C = (0 + 3) \bmod 26$$

$$C = 3 \bmod 26$$

$$C = 3$$

$$C = D$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R	D	F					

$$C = (p + k) \bmod 26$$

$$C = (2 + 3) \bmod 26$$

$$C = 5 \bmod 26$$

$$C = 5$$

$$C = F$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R	D	F	D				

$$C = (p + k) \bmod 26$$

$$C = (0 + 3) \bmod 26$$

$$C = 3 \bmod 26$$

$$C = 3$$

$$C = D$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R	D	F	D	G			

$$C = (p + k) \bmod 26$$

$$C = (3 + 3) \bmod 26$$

$$C = 6 \bmod 26$$

$$C = 6$$

$$C = G$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R	D	F	D	G	H		

$$C = (p + k) \bmod 26$$

$$C = (4 + 3) \bmod 26$$

$$C = 7 \bmod 26$$

$$C = 7$$

$$C = H$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R	D	F	D	G	H	P	

$$C = (p + k) \bmod 26$$

$$C = (12 + 3) \bmod 26$$

$$C = 15 \bmod 26$$

$$C = 15$$

$$C = P$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R	D	F	D	G	H	P	B

$$C = (p + k) \bmod 26$$

$$C = (24 + 3) \bmod 26$$

$$C = 27 \bmod 26$$

$$C = 1$$

$$C = B$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R	D	F	D	G	H	P	B

Plaintext: nesoacademy

Ciphertext: QHVRDFDGHPB

Homework!

Decrypt "QHWZRUN VHFXULWB" using Caesar cipher.





Cryptography and Network Security

Caesar Cipher (Part 2)

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Know about shift ciphers.
- ★ Know the pros and cons of Caesar cipher.
- ★ Understand Brute force attack on Caesar cipher.

Classical Encryption Techniques

1. Substitution Technique
2. Transposition Technique

Classical Encryption Technique

Substitution

- ❖ Caesar Cipher ←
- ❖ Monoalphabetic Substitution
- ❖ Playfair Cipher
- ❖ Hill Cipher
- ❖ Polyalphabetic Ciphers
- ❖ One-Time Pad

Transposition

- ❖ Rail Fence
- ❖ Row Column Transposition

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R	D	F	D	G	H	P	B

Plaintext: nesoacademy

Ciphertext: QHVRDFDGHPB

Shift Cipher

Key = 2, 3, 4, 5, ...

Shift Cipher with Key = 3 is called Caesar Cipher.

Example:

Plaintext : Neso

Key : 4

Ciphertext: RIWS

O	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Pros and Cons

Pros

1. Simple
2. Easy to implement.

Cons

1. The encryption and decryption algorithms are known.
2. There are only 25 keys to try. (Vulnerable to Brute-force attack)
3. The language of the plaintext is known and easily recognizable.

Brute force attack

Ciphertext: SQDYMZK

Shifts	Back	Result
0	[26]	SQDYMZK
1	[25]	TREZNAL
2	[24]	USFAOBM
3	[23]	VTGBPCN
4	[22]	WUHCQDO
5	[21]	XVIDREP
6	[20]	YWJESFQ
7	[19]	ZXKFTGR
8	[18]	AYLGUHS
9	[17]	BZMHVIT
10	[16]	CANIWJU
11	[15]	DBOJXKV
12	[14]	ECPKYLW
13	[13]	FDQLZMX

Shifts	Back	Result
13	[13]	FDQLZMX
14	[12]	GERMANY
15	[11]	HFSNBOZ
16	[10]	IGTOCPA
17	[9]	JHUPDQB
18	[8]	KIVQERC
19	[7]	LJWRFSO
20	[6]	MKXSGTE
21	[5]	NLYTHUF
22	[4]	OMZUIVG
23	[3]	PNAVJWH
24	[2]	QOBWKXI
25	[1]	RPCXLYJ

Homework!

1. Encrypt the plaintext "cryptography" using shift cipher (Key=4).
2. Encrypt "Neso" using shift cipher (Key=0).



Cryptography and Network Security

Monoalphabetic Ciphers

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Recall Caesar cipher.
- ★ Understand permutation.
- ★ Understand Monoalphabetic ciphers with example.
- ★ Know the pros and cons of Monoalphabetic ciphers.

Classical Encryption Techniques

1. Substitution Technique
2. Transposition Technique

Classical Encryption Technique

Substitution

- ❖ Caesar Cipher
- ❖ Monoalphabetic Cipher
- ❖ Playfair Cipher
- ❖ Hill Cipher
- ❖ Polyalphabetic Ciphers
- ❖ One-Time Pad

Transposition

- ❖ Rail Fence
- ❖ Row Column Transposition

Caesar Cipher

- ★ Letters are replaced by other letters or symbols.
- ★ The earlier known and simplest method used by Julius Caesar.
- ★ Replacing each letter of the alphabet with the letter standing three places further down the alphabet.

Example:

O	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Monoalphabetic Cipher

- ★ The “cipher” line can be any permutation of the 26 alphabetic characters.
- ★ This would seem to eliminate brute-force techniques for cryptanalysis.
- ★ A single cipher alphabet (mapping from plain alphabet to cipher alphabet) is used per message.
- ★ English language - Nature of plain text is known.

Permutation

- ★ A permutation of a finite set of elements 'S' is an ordered sequence of all the elements of 'S', with each element appearing exactly once.
- ★ For example $S = \{ a , b , c \}$
- ★ There are six permutations of S.

abc

acb

bac

bca

cab

cba

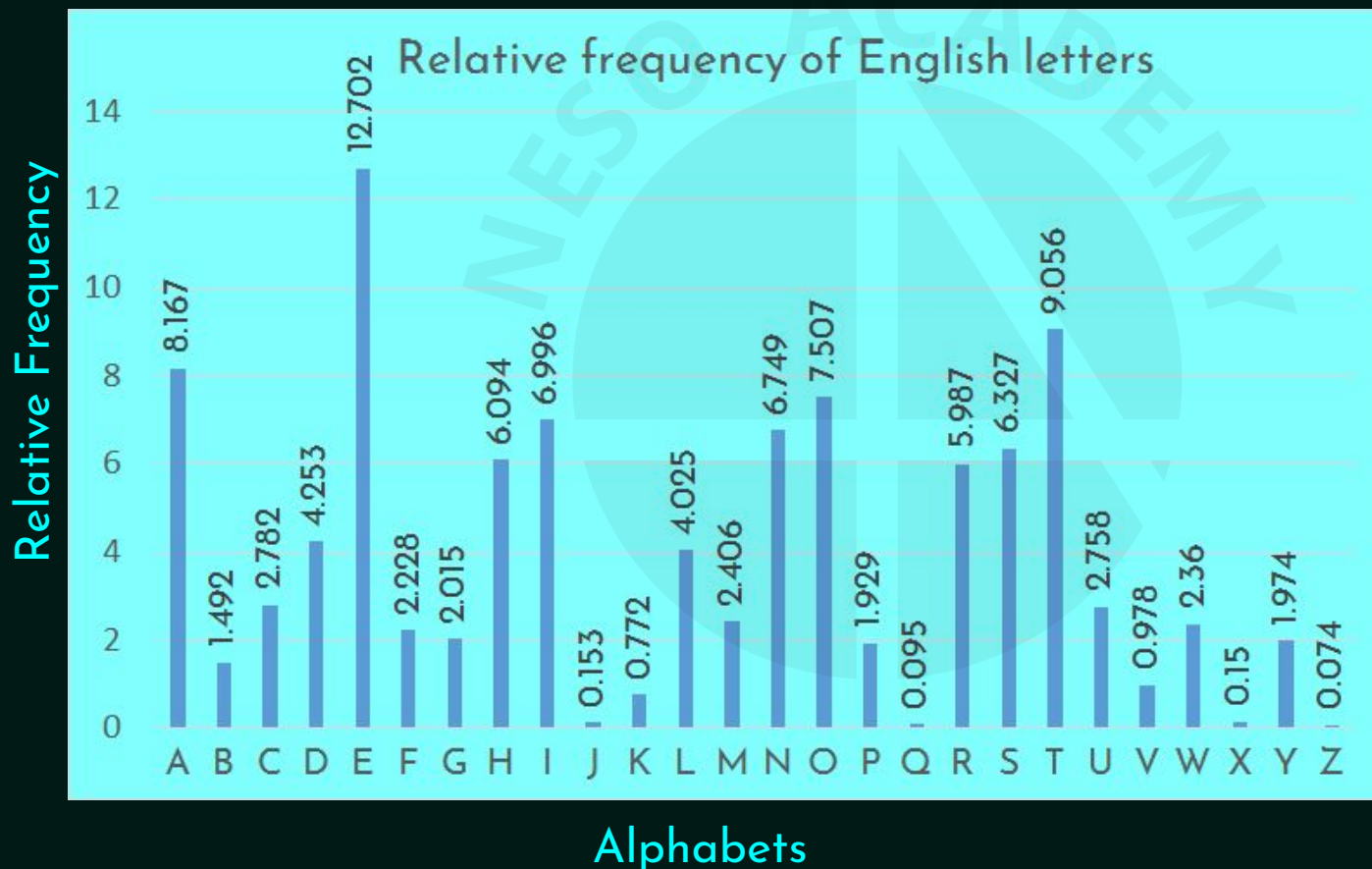
Monoalphabetic Cipher

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Example: Plain Text - Neso

Cipher Text - DULA

Relative frequency of English letters



Monoalphabetic Cipher – Example

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

Monoalphabetic Cipher – Example

P 13.33

E 5.00

B 1.67

N 0.00

Z 11.67

V 4.17

G 1.67

R 0.00

S 8.33

X 4.17

Y 1.67

U 8.33

F 3.33

I 0.83

O 7.50

W 3.33

J 0.83

M 6.67

Q 2.50

C 0.00

H 5.83

T 2.50

K 0.00

D 5.00

A 1.67

L 0.00

Monoalphabetic Cipher – Example

GZGEWVGRNCP

CT	G	Z	G	E	W	V	G	R	N	C	P
PT	E		E				E				
PT	E		E			T	E				
PT	E		E			T	E			A	
PT	E		E			T	E		L	A	N
PT	E		E			T	E	P	L	A	N
PT	E	X	E	C	U	T	E	P	L	A	N

Monoalphabetic Cipher – Example

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
t a e e te a t h a t e e a a
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
e t ta t ha e ee a e th t a
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ
e e e tat e the t

Monoalphabetic Cipher – Example

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

it was disclosed yesterday that several informal but
direct contacts have been made with political
representatives of the viet cong in moscow

Monoalphabetic Cipher – Pros and Cons

Pros

1. Better security than Caesar cipher.

Cons

1. Monoalphabetic ciphers are easy to break because they reflect the frequency data of the original alphabet.
2. Prone to guessing attack using the English letter frequency of occurrence of letters.
3. A countermeasure is to provide multiple substitutes, known as homophones, for a single letter.

Homework!

Encrypt the plaintext “Attack postponed to tomorrow and do not use our secret paper until further info” using Monoalphabetic cipher technique.

Secret Key: The quick brown fox jumps over the lazy dog.

Note: Ignore the second and latter occurrence of alphabets in the key.



Cryptography and Network Security

Playfair Cipher (Part 1)

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand the basic working of playfair cipher.
- ★ Know about digrams and generation of digrams with/without filler character.

Classical Encryption Techniques

1. Substitution Technique
2. Transposition Technique

Classical Encryption Technique

Substitution

- ❖ Caesar Cipher
- ❖ Monoalphabetic Cipher
- ❖ Playfair Cipher
- ❖ Hill Cipher
- ❖ Polyalphabetic Ciphers
- ❖ One-Time Pad

Transposition

- ❖ Rail Fence
- ❖ Row Column Transposition

Playfair Cipher

- ★ Aka Playfair square or Wheatstone-Playfair cipher.
- ★ Manual symmetric encryption technique.
- ★ The first literal digram substitution cipher.
- ★ Invented in 1854 by Charles Wheatstone.
- ★ Bore the name of Lord Playfair for promoting its use.

Playfair Cipher

- ★ Multiple letter encryption cipher.
- ★ Digrams.
- ★ 5 x 5 matrix constructed using a keyword (Ex: Monarchy)

Playfair Cipher

- ★ Multiple letter encryption cipher.
- ★ Digrams.
- ★ 5 x 5 matrix constructed using a keyword (Ex: Monarchy)

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Rules for encryption using Playfair Cipher

1. Digrams.
2. Repeating Letters - Filler letter.
3. Same Column | ↓ | Wrap around.
4. Same row | → | Wrap around.
5. Rectangle | ⇔ | Swap

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Example

Plaintext: attack

Digrams: at ta ck

Plaintext: neso academy

Digrams: ne so ac ad em yx

Plaintext: balloon

Digrams: ba ll oo n

Digrams: ba lx lo on

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z



Cryptography and Network Security

Playfair Cipher (Part 2)

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Recall the rules for encryption using Playfair ciphering technique.
- ★ Generate ciphertext using Playfair ciphering technique.

Playfair Cipher

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Rules for encryption using Playfair Cipher

1. Digrams.
2. Repeating Letters - Filler letter.
3. Same Column | $\downarrow$ | Wrap around.
4. Same row | $\rightarrow$ | Wrap around.
5. Rectangle | $\Leftrightarrow$ | Swap

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Understanding the rules

Example 1: attack

Digrams: at ta ck

at	ta	ck
RS	SR	DE

Plaintext: attack

Digrams: RSSRDE

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Understanding the rules

Example 2: mosque

mo	sq	ue
ON	TS	ML

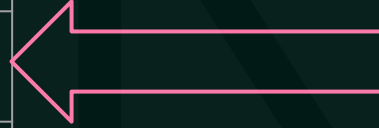
Plaintext: mosque

Digrams: ONTSML

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Rapid Fire – Activity

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z



Digrams

oa

ab

gk

dc

rz

zr

mz

rm



Cryptography and Network Security

Playfair Cipher (Solved Question)

Question

Encrypt the message "Hide the gold under the carpet" using playfair technique with the keyword "Neso Academy".

Question

Encrypt the message "Hide the gold under the carpet" using playfair technique with the keyword "Neso Academy".

Plaintext : Hide the gold under the carpet

Key : Neso Academy

Ciphertext : IKGDQKDPNRCVECOPOKNDOTVDRZ

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

Homework

Decrypt the ciphertext “ODZFAQSEZSONTSW” using playfair technique with the keyword “Neso App”.

Answer: You are awesome.



Cryptography and Network Security

Hill Cipher (Encryption)

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand the basic working of Hill cipher.
- ★ Understand the encryption process in Hill cipher.
- ★ Know how to generate the ciphertext using Hill cipher.

Classical Encryption Techniques

1. Substitution Technique
2. Transposition Technique

Classical Encryption Technique

Substitution

- ❖ Caesar Cipher
- ❖ Monoalphabetic Cipher
- ❖ Playfair Cipher
- ❖ Hill Cipher
- ❖ Polyalphabetic Ciphers
- ❖ One-Time Pad

Transposition

- ❖ Rail Fence
- ❖ Row Column Transposition

Hill Cipher

- ★ Multi-letter cipher.
- ★ Developed by Lester Hill in 1929.
- ★ Encrypts a group of letters : digraph, trigraph or polygraph.

Hill Cipher

- ★ Review few terminologies from linear algebra.
- ★ Concepts to be known:
 - Matrix arithmetic modulo 26.
 - Square matrix.
 - Determinant.
 - Multiplicative inverse.

The Hill Algorithm

This can be expressed as

$$C = E(K,P) = P \times K \bmod 26$$

$$P = D(K,C) = C K^{-1} \bmod 26 = P \times K \times K^{-1} \bmod 26$$

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$


$$C_1 = (P_1 K_{11} + P_2 K_{21} + P_3 K_{31}) \bmod 26$$

$$C_2 = (P_1 K_{12} + P_2 K_{22} + P_3 K_{32}) \bmod 26$$

$$C_3 = (P_1 K_{13} + P_2 K_{23} + P_3 K_{33}) \bmod 26$$

Hill Cipher Example

Question: Encrypt “pay more money” using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Solution:

p	a	y	m	o	r	e	m	o	n	e	y
15	0	24	12	14	17	4	12	14	13	4	24

Key = 3 x 3 matrix.

PT = pay mor emo ney

Hill Cipher Example

Encrypting: pay

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \text{mod } 26$$

$$(C_1 \ C_2 \ C_3) = (15 \ 0 \ 24) \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \text{mod } 26$$

$$= (15 \times 17 + 0 \times 21 + 24 \times 2 \quad 15 \times 17 + 0 \times 18 + 24 \times 2 \quad 15 \times 5 + 0 \times 21 + 24 \times 19) \text{mod } 26$$

$$= (303 \ 303 \ 531) \text{mod } 26$$

$$= (17 \ 17 \ 11)$$

$$= (R \ R \ L)$$

Hill Cipher Example

Encrypting: mor

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \text{mod } 26$$

$$(C_1 \ C_2 \ C_3) = (12 \ 14 \ 17) \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \text{mod } 26$$

$$= (12 \times 17 + 14 \times 21 + 17 \times 2 \quad 12 \times 17 + 14 \times 18 + 17 \times 2 \quad 12 \times 5 + 14 \times 21 + 17 \times 19) \text{mod } 26$$

$$= (532 \ 490 \ 677) \text{mod } 26$$

$$= (12 \ 22 \ 1)$$

$$= (M \ W \ B)$$

Hill Cipher Example

PT	p	a	y	m	o	r	e	m	o	n	e	y
CT	R	R	L	M	W	B						

Hill Cipher Example

Encrypting: emo

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \text{ mod } 26$$

$$(C_1 \ C_2 \ C_3) = (4 \ 12 \ 14) \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \text{ mod } 26$$

$$= (4 \times 17 + 12 \times 21 + 14 \times 2 \quad 4 \times 17 + 12 \times 18 + 14 \times 2 \quad 4 \times 5 + 12 \times 21 + 14 \times 19) \text{ mod } 26$$

$$= (348 \ 312 \ 538) \text{ mod } 26$$

$$= (10 \ 0 \ 18)$$

$$= (K \ A \ S)$$

Hill Cipher Example

PT	p	a	y	m	o	r	e	m	o	n	e	y
CT	R	R	L	M	W	B	K	A	S			

Hill Cipher Example

Encrypting: ney

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \text{ mod } 26$$

$$\begin{aligned} (C_1 \ C_2 \ C_3) &= (13 \ 4 \ 24) \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \text{ mod } 26 \\ &= (13 \times 17 + 4 \times 21 + 24 \times 2 \quad 13 \times 17 + 4 \times 18 + 24 \times 2 \quad 13 \times 5 + 4 \times 21 + 24 \times 19) \text{ mod } 26 \\ &= (348 \ 312 \ 538) \text{ mod } 26 \\ &= (15 \ 3 \ 7) \\ &= (P \ D \ H) \end{aligned}$$

Hill Cipher Example

PT	p	a	y	m	o	r	e	m	o	n	e	y
CT	R	R	L	M	W	B	K	A	S	P	D	H

Hill Cipher Example

Question: Encrypt “pay more money” using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Plaintext : pay more money

Ciphertext : RRLMWBKASPDH



Cryptography and Network Security

Hill Cipher (Decryption)

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand the basic working of Hill cipher.
- ★ Understand the decryption process in Hill cipher.
- ★ Know how to decrypt the ciphertext using Hill cipher.

Classical Encryption Techniques

1. Substitution Technique

2. Transposition Technique

Classical Encryption Technique

Substitution

- ❖ Caesar Cipher
- ❖ Monoalphabetic Cipher
- ❖ Playfair Cipher
- ❖ Hill Cipher
- ❖ Polyalphabetic Ciphers
- ❖ One-Time Pad

Transposition

- ❖ Rail Fence
- ❖ Row Column Transposition

Hill Cipher

- ★ Review few terminologies from linear algebra.
- ★ Concepts to be known:
 - Matrix arithmetic modulo 26.
 - Square matrix.
 - Determinant.
 - Adjoint matrix.
 - Multiplicative inverse.

The Hill Algorithm

This can be expressed as

Encryption:

$$C = E(K,P) = P \times K \bmod 26$$

Decryption:

$$P = D(K,C) = C \times K^{-1} \bmod 26$$

The Hill Algorithm

This can be expressed as

$$C = E(K,P) = P \times K \text{ mod } 26$$

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \text{ mod } 26$$


$$C_1 = (P_1 K_{11} + P_2 K_{21} + P_3 K_{31}) \text{ mod } 26$$

$$C_2 = (P_1 K_{12} + P_2 K_{22} + P_3 K_{32}) \text{ mod } 26$$

$$C_3 = (P_1 K_{13} + P_2 K_{23} + P_3 K_{33}) \text{ mod } 26$$

The Hill Algorithm

This can be expressed as

Encryption:

$$C = E(K,P) = P \times K \bmod 26$$

Decryption:

$$P = D(K,C) = C \times K^{-1} \bmod 26$$

Decryption requires K^{-1} , the inverse matrix K .

The Hill Algorithm

To find the determinant of K: $\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$

$$\text{Det} \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$= 17(18 \times 19 - 2 \times 21) - 17(19 \times 21 - 2 \times 21) + 5(2 \times 21 - 2 \times 18) \bmod 26$$

$$= 17(342 - 42) - 17(399 - 42) + 5(42 - 36) \bmod 26$$

$$= 17(300) - 17(357) + 5(6) \bmod 26$$

$$= 5100 - 6069 + 30 \bmod 26$$

$$= -939 \bmod 26$$

$$= -3 \bmod 26$$

$$= 23$$

The Hill Algorithm

$$K^{-1} = \frac{1}{\text{Det } K} \times \text{Adjoint } K$$

To find Adjoint K

$$\text{Adj } K = \begin{vmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{vmatrix}$$

$$\text{Adj } K = \begin{vmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{vmatrix}$$

$$\text{Adj } K = \begin{vmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{vmatrix}$$

The Hill Algorithm

Adj K =

17	17	5	17	17
21	18	21	21	18
2	2	19	2	2

Adj K =

17	17	5	17	17
21	18	21	21	18
2	2	19	2	2

Adj K =

17	17	5	17	17
21	18	21	21	18
2	2	19	2	2
17	17	5	17	17
21	18	21	21	18

The Hill Algorithm

Adj K =

17	17	5	17	17
21	18	21	21	18
2	2	19	2	2

17	17	5	17	17
21	18	21	21	18

Adj K =

17	17	5	17	17
21	18	21	21	18
2	2	19	2	2

17	17	5	17	17
21	18	21	21	18

The Hill Algorithm

A 4x4 grid of numbers with arrows indicating a path from the top-left to the bottom-right. The path consists of teal numbers: 18, 21, 21, 18 in the top row; 2, 19, 2, 2 in the second row; 17, 5, 17, 17 in the third row; and 18, 21, 21, 18 in the bottom row. Orange arrows show the path moving right and down, with some arrows crossing each other.

Performing the operation - Column wise

Entering the matrix - Row wise

$$\begin{aligned}
 &= \begin{array}{ccc} 18 \times 19 - 2 \times 21 & 2 \times 5 - 17 \times 19 & 17 \times 21 - 18 \times 5 \\ 21 \times 2 - 19 \times 21 & 19 \times 17 - 5 \times 2 & 5 \times 21 - 21 \times 17 \\ 21 \times 2 - 2 \times 18 & 2 \times 17 - 17 \times 2 & 17 \times 18 - 21 \times 17 \end{array} \\
 &= \begin{array}{ccc} 300 & -313 & 267 \\ -357 & 313 & -252 \text{ mod } 26 \\ 6 & 0 & -51 \end{array}
 \end{aligned}$$

The Hill Algorithm



Performing the operation - Column wise

Entering the matrix - Row wise

$$\begin{aligned}
 &= \begin{matrix} 18 \times 19 - 2 \times 21 & 2 \times 5 - 17 \times 19 & 17 \times 21 - 18 \times 5 \\ 21 \times 2 - 19 \times 21 & 19 \times 17 - 5 \times 2 & 5 \times 21 - 21 \times 17 \\ 21 \times 2 - 2 \times 18 & 2 \times 17 - 17 \times 2 & 17 \times 18 - 21 \times 17 \end{matrix} \\
 &= \begin{matrix} 14 & -1 & 7 \\ -19 & 1 & -18 \\ 6 & 0 & -25 \end{matrix} \pmod{26}
 \end{aligned}$$

The Hill Algorithm



Performing the operation - Column wise

Entering the matrix - Row wise

$$= \begin{matrix} 18 \times 19 - 2 \times 21 & 2 \times 5 - 17 \times 19 & 17 \times 21 - 18 \times 5 \\ 21 \times 2 - 19 \times 21 & 19 \times 17 - 5 \times 2 & 5 \times 21 - 21 \times 17 \\ 21 \times 2 - 2 \times 18 & 2 \times 17 - 17 \times 2 & 17 \times 18 - 21 \times 17 \end{matrix}$$

$$= \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \bmod 26$$

The Hill Algorithm

Decryption requires K^{-1} , the inverse matrix K .

$$K^{-1} = \frac{1}{\text{Det } K} \times \text{Adj } K$$

$$K^{-1} = \frac{1}{23} \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \text{ mod } 26$$

$$K^{-1} = 23^{-1} \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \text{ mod } 26$$

The Hill Algorithm

Decryption requires K^{-1} , the inverse matrix K .

$$K^{-1} = \frac{1}{\text{Det } K} \times \text{Adj } K$$

$$K^{-1} = \frac{1}{23} \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \text{ mod } 26$$

$$K^{-1} = 23^{-1} \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \text{ mod } 26$$

$23^{-1} \times 23 = 1 \text{ mod } 26$	$9 \times 23 = 25 \text{ mod } 26$
$1 \times 23 = 23 \text{ mod } 26$	$10 \times 23 = 22 \text{ mod } 26$
$2 \times 23 = 20 \text{ mod } 26$	$11 \times 23 = 19 \text{ mod } 26$
$3 \times 23 = 17 \text{ mod } 26$	$12 \times 23 = 16 \text{ mod } 26$
$4 \times 23 = 14 \text{ mod } 26$	$13 \times 23 = 13 \text{ mod } 26$
$5 \times 23 = 11 \text{ mod } 26$	$14 \times 23 = 10 \text{ mod } 26$
$6 \times 23 = 8 \text{ mod } 26$	$15 \times 23 = 7 \text{ mod } 26$
$7 \times 23 = 5 \text{ mod } 26$	$16 \times 23 = 4 \text{ mod } 26$
$8 \times 23 = 2 \text{ mod } 26$	$17 \times 23 = 1 \text{ mod } 26$

The Hill Algorithm

Decryption requires K^{-1} , the inverse matrix K .

$$K^{-1} = \frac{1}{\text{Det } K} \times \text{Adj } K$$

$$K^{-1} = \frac{1}{23} \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \text{ mod } 26$$

$$K^{-1} = 17 \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \text{ mod } 26$$

The Hill Algorithm

$$K^{-1} = 17 \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \bmod 26$$

$$K^{-1} = \begin{pmatrix} 238 & 425 & 119 \\ 119 & 17 & 136 \\ 102 & 0 & 17 \end{pmatrix} \bmod 26$$

$$K^{-1} = \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix}$$

The Hill Algorithm

$$K \times K^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$K = \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \quad K^{-1} = \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix}$$

This is demonstrated as

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} = \begin{pmatrix} 443 & 442 & 442 \\ 858 & 495 & 780 \\ 494 & 52 & 365 \end{pmatrix} \bmod 26$$
$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Hill Cipher Example

Question: Encrypt “pay more money” using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Plaintext : pay more money

Ciphertext : RRLMWBKASPDH

Hill Cipher Example

Question: Decrypt "RRLMWBKASPDH" using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Solution:

$$P = C K^{-1} \bmod 26$$

R	R	L	M	W	B	K	A	S	P	D	H
17	17	11	12	22	1	10	0	18	15	3	7

Hill Cipher Example

Decrypting: RRL

$$(P_1 \ P_2 \ P_3) = (R \ R \ L) \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \text{ mod } 26$$


$$\begin{aligned} (C_1 \ C_2 \ C_3) &= (17 \ 17 \ 14) \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \text{ mod } 26 \\ &= (17 \times 4 + 17 \times 15 + 14 \times 24 \quad 17 \times 9 + 17 \times 17 + 14 \times 0 \quad 17 \times 15 + 17 \times 6 + 14 \times 17) \text{ mod } 26 \\ &= (587 \ 442 \ 544) \text{ mod } 26 \\ &= (15 \ 0 \ 24) \\ &= (P \ A \ Y) \end{aligned}$$

Hill Cipher Example

Decrypting: MWB

$$(P_1 \ P_2 \ P_3) = (M \ W \ B) \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \text{ mod } 26$$

$$(C_1 \ C_2 \ C_3) = (12 \ 22 \ 1) \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \text{ mod } 26$$

$$= (12 \times 4 + 22 \times 15 + 1 \times 24 \quad 12 \times 9 + 22 \times 17 + 1 \times 0 \quad 12 \times 15 + 22 \times 6 + 1 \times 17) \text{ mod } 26$$

$$= (402 \ 482 \ 329) \text{ mod } 26$$

$$= (12 \ 14 \ 17)$$

$$= (M \ O \ R)$$

Hill Cipher Example

CT	R	R	L	M	W	B	K	A	S	P	D	H
PT	p	a	y	m	o	r						

Hill Cipher Example

Decrypting: KAS

$$(P_1 \ P_2 \ P_3) = (K \ A \ S) \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \text{ mod } 26$$

$$\begin{aligned} (C_1 \ C_2 \ C_3) &= (10 \ 0 \ 18) \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \text{ mod } 26 \\ &= (10 \times 4 + 0 \times 15 + 18 \times 24 \quad 10 \times 9 + 0 \times 17 + 18 \times 0 \quad 10 \times 15 + 0 \times 6 + 18 \times 17) \text{ mod } 26 \\ &= (472 \ 90 \ 456) \text{ mod } 26 \\ &= (4 \ 12 \ 14) \\ &= (E \ M \ O) \end{aligned}$$

Hill Cipher Example

CT	R	R	L	M	W	B	K	A	S	P	D	H
PT	p	a	y	m	o	r	e	m	o			

Hill Cipher Example

Decrypting: PDH

$$(P_1 \ P_2 \ P_3) = (P \ D \ H) \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \text{mod } 26$$

$$\begin{aligned} (C_1 \ C_2 \ C_3) &= (15 \ 3 \ 7) \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \text{mod } 26 \\ &= (15 \times 4 + 3 \times 15 + 7 \times 24 \quad 15 \times 9 + 3 \times 17 + 7 \times 0 \quad 15 \times 15 + 3 \times 6 + 7 \times 17) \text{mod } 26 \\ &= (273 \ 186 \ 362) \text{mod } 26 \\ &= (13 \ 4 \ 24) \\ &= (N \ E \ Y) \end{aligned}$$

Hill Cipher Example

CT	R	R	L	M	W	B	K	A	S	P	D	H
PT	p	a	y	m	o	r	e	m	o	n	e	y

Hill Cipher Example

Question: Decrypt "RRLMWBKASPDH" using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Ciphertext : RRLMWBKASPDH

Plaintext : pay more money



Cryptography and Network Security

Polyalphabetic Ciphers
(Vigenère Cipher)

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand Polyalphabetic substitution cipher.
- ★ Understand the working of Vigenere cipher.
- ★ Understand cryptanalysis on Vigenere cipher.
- ★ Know about autokey system.

Classical Encryption Techniques

1. Substitution Technique
2. Transposition Technique

Classical Encryption Technique

Substitution

- ❖ Caesar Cipher
- ❖ Monoalphabetic Cipher
- ❖ Playfair Cipher
- ❖ Hill Cipher
- ❖ Polyalphabetic Ciphers
- ❖ One-Time Pad

Transposition

- ❖ Rail Fence
- ❖ Row Column Transposition

Polyalphabetic Cipher

- ★ To improve on the simple monoalphabetic technique.
- ★ General name: Polyalphabetic substitution cipher.

Common features

1. A set of related monoalphabetic substitution rules is used.
2. A key determines which particular rule is chosen for a given transformation.

Vigenere Cipher

- ★ It consists of the 26 Caesar ciphers with shifts of 0 through 25.

Encryption process:

$$C_i = (P_i + K_{i \bmod m}) \bmod 26$$

Decryption process:

$$P_i = (C_i - K_{i \bmod m}) \bmod 26$$

Vigenere Cipher

Key : deceptivedeceptivedeceptive

Plaintext : wearediscoveredsaveyourself

Ciphertext : ZICVTWQNGRZGVTWAVZHCOYGLMGJ

Key	3	4	2	4	15	19	8	21	4	3	4	2	4
PT	22	4	0	17	4	3	8	18	2	14	21	4	17
CT	25	8	2	21	19	22	16	13	6	17	25	6	21

Key	15	19	8	21	4	3	4	2	4	15	19	8	21	4
PT	4	3	18	0	21	4	24	14	20	17	18	4	11	5
CT	19	22	0	21	25	7	2	16	24	6	11	12	6	9

Vigenere Cipher – Cryptanalysis

- ★ Determining the length of the keyword.
- ★ Key and the plaintext share the same frequency distribution of letters, a statistical technique can be applied.

Autokey system

- ★ The periodic nature of the keyword can be eliminated by using a non-repeating keyword that is as long as the message itself.
- ★ Vigenère proposed autokey system, in which a keyword is concatenated with the plaintext itself to provide a running key.

Example

Key : deceptivewearediscoveredsav

Plaintext : wearediscoveredsaveyourself

Ciphertext : ZICVTWQNGKZEIIGASXSTSLVWLA



Cryptography and Network Security

Polyalphabetic Cipher
(Vernam Cipher)

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand Polyalphabetic substitution cipher.
- ★ Understand the working of Vernam cipher.
- ★ Know the cryptanalytic difficulty of Vernam cipher.

Classical Encryption Techniques

1. Substitution Technique
2. Transposition Technique

Classical Encryption Technique

Substitution

- ❖ Caesar Cipher
- ❖ Monoalphabetic Cipher
- ❖ Playfair Cipher
- ❖ Hill Cipher
- ❖ Polyalphabetic Ciphers
- ❖ One-Time Pad

Transposition

- ❖ Rail Fence
- ❖ Row Column Transposition

Vernam Cipher

- ★ Need of ultimate defense against cryptanalytic attack.
- ★ Length of the keyword = Length of the plaintext.
- ★ No statistical relationship to it.
- ★ AT&T engineer named Gilbert Vernam in 1918.
- ★ His system works on binary bits rather than letter.
- ★ The system can be expressed as follows:

$$C_i = P_i \oplus K_i$$

Vernam Cipher

$$C_i = P_i \oplus K_i$$

where

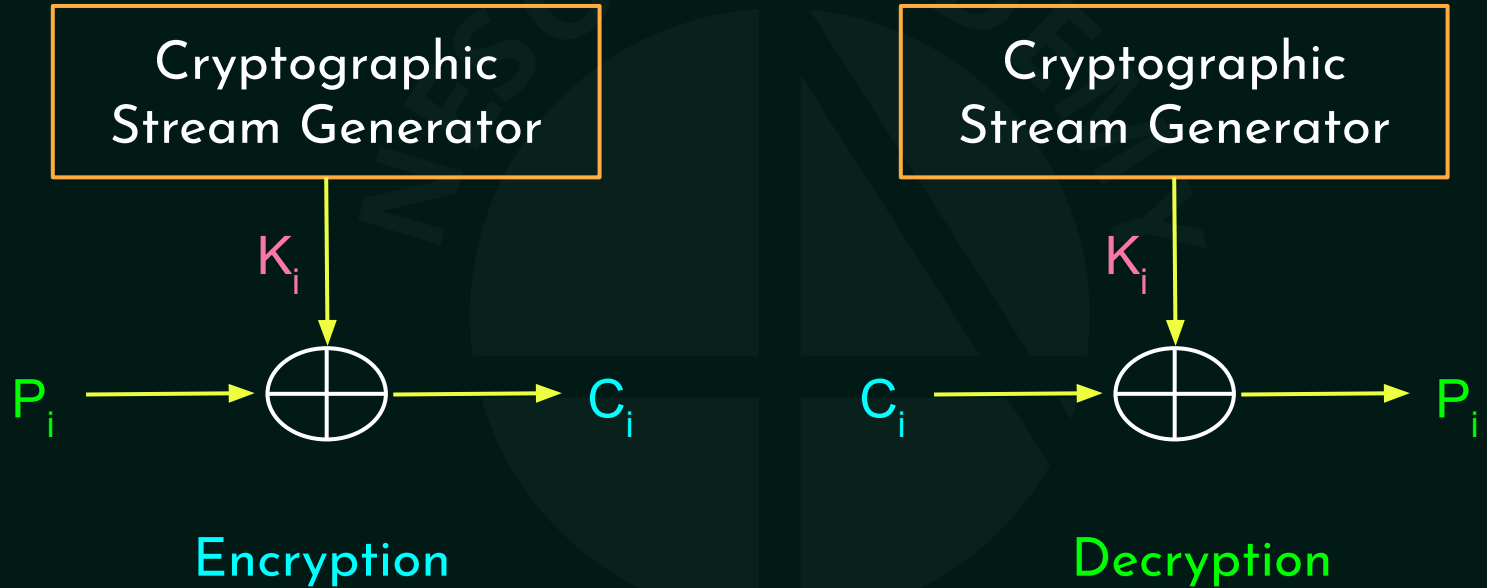
P_i = i^{th} binary of the plaintext.

K_i = i^{th} binary of the key.

C_i = i^{th} binary of the ciphertext.

$\oplus$ = XOR operation.

Vernam Cipher



Vernam Cipher – Cryptanalysis

- ★ Construction of the key.
- ★ The use of a running loop of tape that eventually repeated the key.
- ★ The system worked with a very long but repeating keyword.
- ★ This technique can be broken with sufficient ciphertext, the use of known or probable plaintext sequences, or both.



Cryptography and Network Security

One Time Pad

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand One Time Pad encryption technique.
- ★ Know the two fundamental difficulties of One Time Pad encryption technique.
- ★ Know about Perfect secrecy.

Classical Encryption Techniques

1. Substitution Technique
2. Transposition Technique

Classical Encryption Technique

Substitution

- ❖ Caesar Cipher
- ❖ Monoalphabetic Cipher
- ❖ Playfair Cipher
- ❖ Hill Cipher
- ❖ Polyalphabetic Ciphers
- ❖ One-Time Pad

Transposition

- ❖ Rail Fence
- ❖ Row Column Transposition

One Time Pad

- ★ Improvement to the Vernam cipher.
- ★ It yields the ultimate in security.
- ★ Random key that is as long as the message.
- ★ The key need not be repeated.
- ★ In addition, the key is to be used to encrypt and decrypt a single message, and then is discarded.
- ★ Each new message requires a new key of the same length as the new message.
- ★ Such a scheme, known as a one-time pad, is unbreakable.

One Time Pad

- ★ It produces random output.
- ★ No statistical relationship to the plaintext.
- ★ Because the ciphertext contains no information whatsoever about the plaintext, there is simply no way to break the code.
- ★ The code is unbreakable.
- ★ The security of the one-time pad is entirely due to the randomness of the key.

Two Fundamental Difficulties

- ★ The practical problem of making large quantities of random keys.
- ★ Even more daunting is the problem of key distribution and protection.
- ★ Because of these difficulties, the one-time pad is of limited utility and is useful primarily for low-bandwidth channels requiring very high security.

Perfect Secrecy

- ★ The one-time pad is the only cryptosystem that exhibits what is referred to as **perfect secrecy**.

Perfect secrecy is the notion that, given an encrypted message (or ciphertext) from a perfectly secure encryption system (or cipher), absolutely nothing will be revealed about the unencrypted message (or plaintext) by the ciphertext.



Cryptography and Network Security

Rail Fence Technique

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand the working of transposition technique.
- ★ Understand the working of Rain Fence technique with example.

Classical Encryption Techniques

1. Substitution Technique

2. Transposition Technique

Classical Encryption Technique

Substitution

- ❖ Caesar Cipher
- ❖ Monoalphabetic Cipher
- ❖ Playfair Cipher
- ❖ Hill Cipher
- ❖ Polyalphabetic Ciphers
- ❖ One-Time Pad

Transposition

- ❖ Rail Fence
- ❖ Row Column Transposition

Transposition Cipher

- ★ Some sort of permutation applied on the plaintext letters.
- ★ This technique is referred to as a transposition cipher.
- ★ The simplest such cipher is the rail fence.

Rail Fence Technique

- ★ The plaintext is written down as a sequence of diagonals and then read off as a sequence of rows.

Example:

Encipher the message “**neso academy is the best**” with a rail fence of depth 2.

Rail Fence Technique

Plaintext : neso academy is the best.

Depth : 2

n		s		a		a		e		y		s		h		b		s	
	e		o		c		d		m		i		t		e		e		t

Ciphertext : NSAAEYSHBSEOCDMITEET

Homework

Plaintext : neso academy is the best.

Depth : 3



Rail Fence

Plaintext : Thank you very much

Depth : 3

T				k				v				m			
	h		n		y		u		e		y		u		h
		a					o				r			c	

Ciphertext: TKVMHNYUEYUHAORE



Cryptography and Network Security

Row Column Transposition

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand the working of row column transposition.
- ★ Generate ciphertext using row column transposition technique.
- ★ Understand how to do more complex permutation that is not easily reconstructed.

Classical Encryption Techniques

1. Substitution Technique

2. Transposition Technique

Classical Encryption Technique

Substitution

- ❖ Caesar Cipher
- ❖ Monoalphabetic Cipher
- ❖ Playfair Cipher
- ❖ Hill Cipher
- ❖ Polyalphabetic Ciphers
- ❖ One-Time Pad

Transposition

- ❖ Rail Fence
- ❖ Row Column Transposition

Row Column Transposition

- ★ A more complex scheme.
- ★ Rectangle.
- ★ Write : Row by row.
- ★ Read : Column by column.
- ★ Key : Order of the column

Row Column Transposition

Example: Encrypt the message "Kill Corona Virus at twelve am tomorrow"

Row Column Transposition

Plaintext : "Kill Corona Virus at twelve am tomorrow"

Key → 4 3 1 2 5 6 7

Plaintext (Input) →

K	i	l	l	c	o	r
o	n	a	v	i	r	u
s	a	t	t	w	e	l
v	e	a	m	t	o	m
o	r	r	o	w	y	z

Ciphertext : LATARLV TMOINA ERKOSV OCITW TWOREO YRULMZ

Row Column Transposition

Plaintext : "Kill Corona Virus at twelve am tomorrow"

Key → 4 3 1 2 5 6 7

Ciphertext (Input) →

L	A	T	A	R	L	V
T	M	O	I	N	A	E
R	K	O	S	V	O	C
I	W	T	W	O	R	E
O	Y	R	U	L	M	Z

Ciphertext : LATARLV TMOINA ERKOSV OC IWTW OREO YRUL MZ



Cryptography and Network Security

Steganography

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand the basics of steganography.
- ★ Know the disadvantages of steganography.
- ★ Understand the difference between cryptography and steganography.

Steganography

- ★ Conceal the existence of the message.
- ★ Hiding the message.
- ★ Not an encryption scheme.
- ★ Cryptography renders the message unintelligible to outsiders by various transformations of the text.
- ★ Example: Simply encrypt correct reading exactly twice.

Secret

Steganography

- ★ Character marking.
- ★ Invisible ink.
- ★ Pin punctures.
- ★ Typewriter color ribbon.

Drawbacks

- ★ Lot of overhead.
- ★ Once the system is discovered, it becomes virtually worthless.

Note:

Alternatively, a message can be first encrypted and then hidden using steganography.

*Tell henry **alice** **not** kane*
*Yesterday **our**plan **undone** !*



Cryptography and Network Security

LSB Steganography – Demo

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Know about LSB steganography.
- ★ Know the difference between watermarking and image steganography.
- ★ Understand practically the operation of LSB steganography with the help of OpenStego tool.

LSB Steganography

★ LSB - Least Significant Bit.

★ 10101001



Least Significant Bits

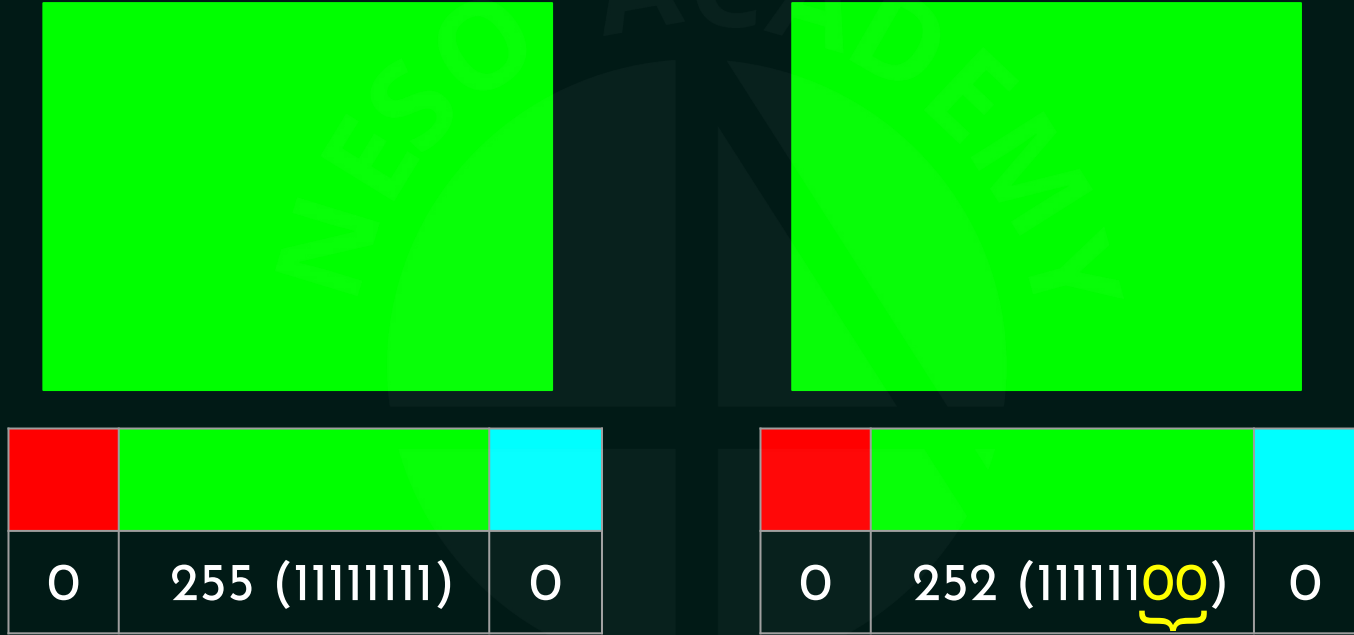
★ RGB color model.

★ Red : RGB(255,0,0) - RGB(11111111, 00000000, 00000000)

★ Green : RGB(0,255,0) - RGB(00000000, 11111111, 00000000)

★ Blue : RGB(0,0,255) - RGB(00000000, 00000000, 11111111)

LSB Steganography



These LSBs can be used for storing the information.

LSB Steganography

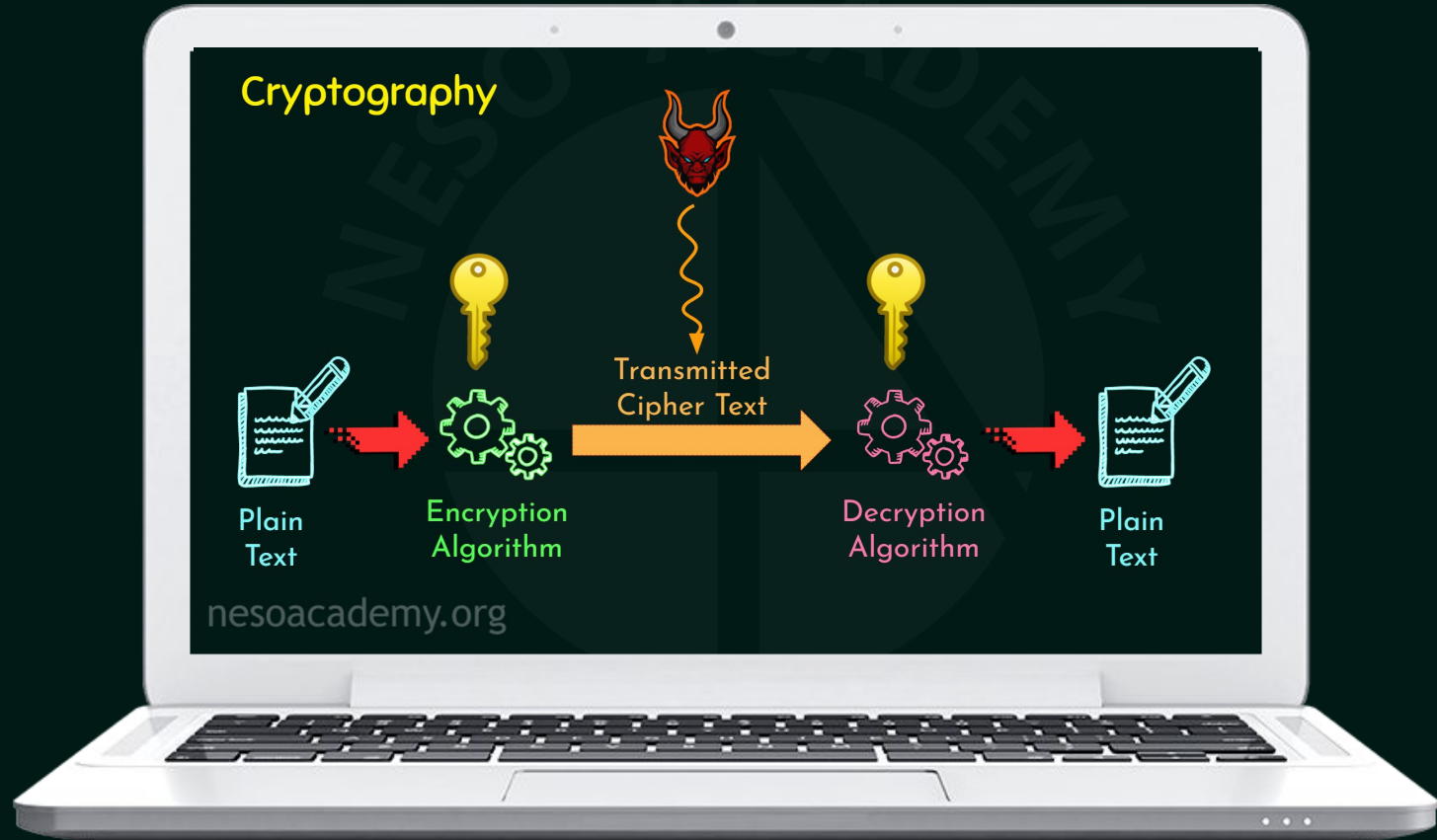


Image without hidden
information



Image with hidden
information

Watermarking





This PC



OpenStego
Demo

All Apps Documents Web More

Best match

Run OpenStego App

Install and Run OpenStego

Apps

Google Chrome

File Explorer

Photos

Settings

Word

Camera

Outlook

Search the web

open - See web results

Folders (6+)

Documents (6+)

Settings (8+)

Photos (1+)

Run OpenStego App

Open

Run as administrator

Open file location

Pin to Start

Pin to taskbar

Uninstall



Create
OpenStego Demo
Folder

Interface to hide the message

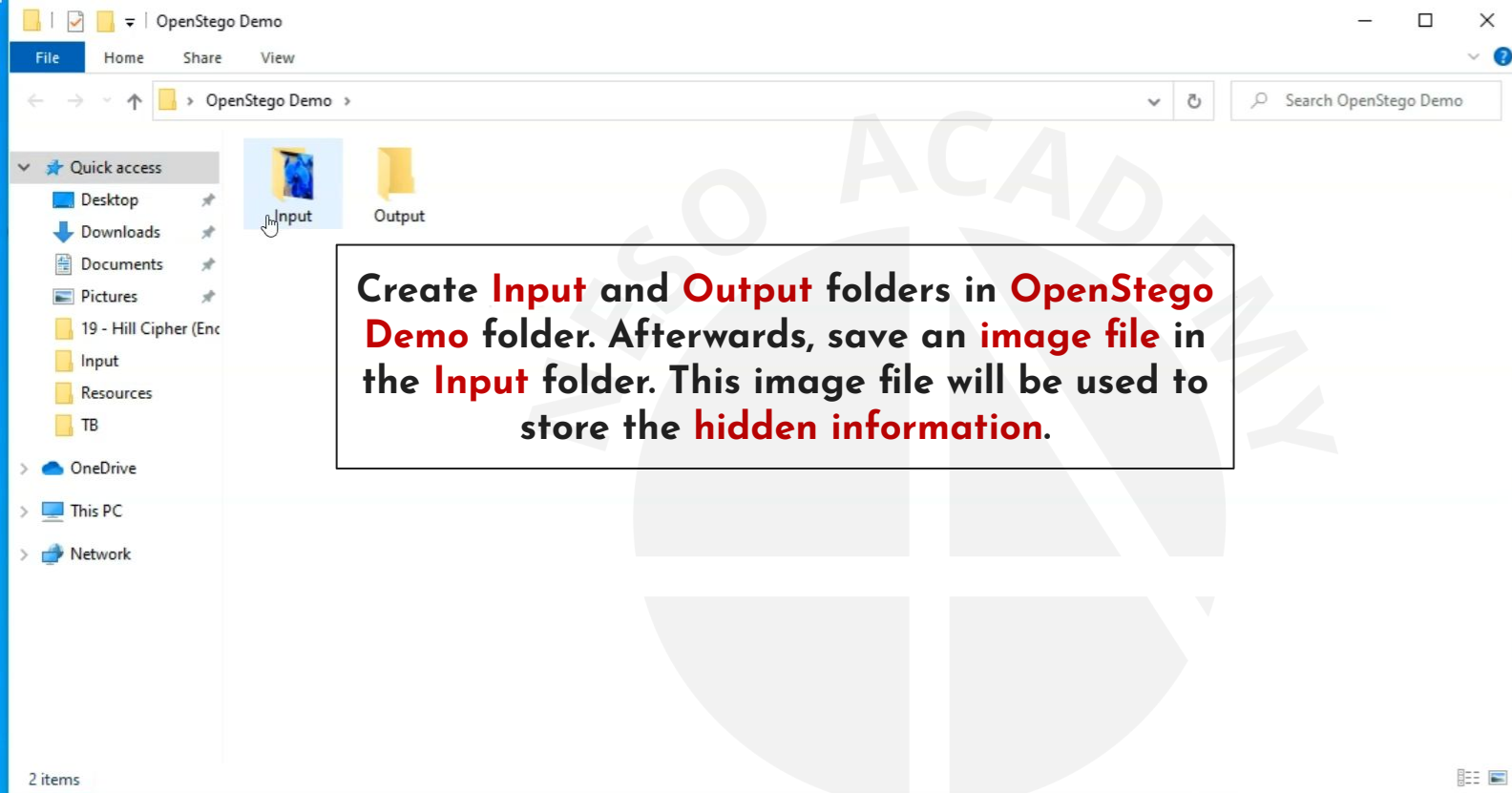
The image shows the OpenStego application window. The title bar reads 'OpenStego'. The menu bar contains 'File' and 'Help'. The main window is divided into two panes. The left pane, titled 'Data Hiding', contains two buttons: 'Hide Data' (highlighted with a blue border) and 'Extract Data'. The right pane, titled 'Hide data in harmless looking files', contains the following fields and controls:

- 'Message File': A text input field with a browse button ('...').
- 'Cover File': A text input field with a browse button ('...'). Below it is the instruction: *(Select multiple files or provide wildcard (*, ?) to embed same message in multiple files)*.
- 'Output Stego File': A text input field with a browse button ('...').
- 'Options' section: Contains 'Password' and 'Confirm Password' text input fields.
- A 'Hide Data' button at the bottom right of the right pane.

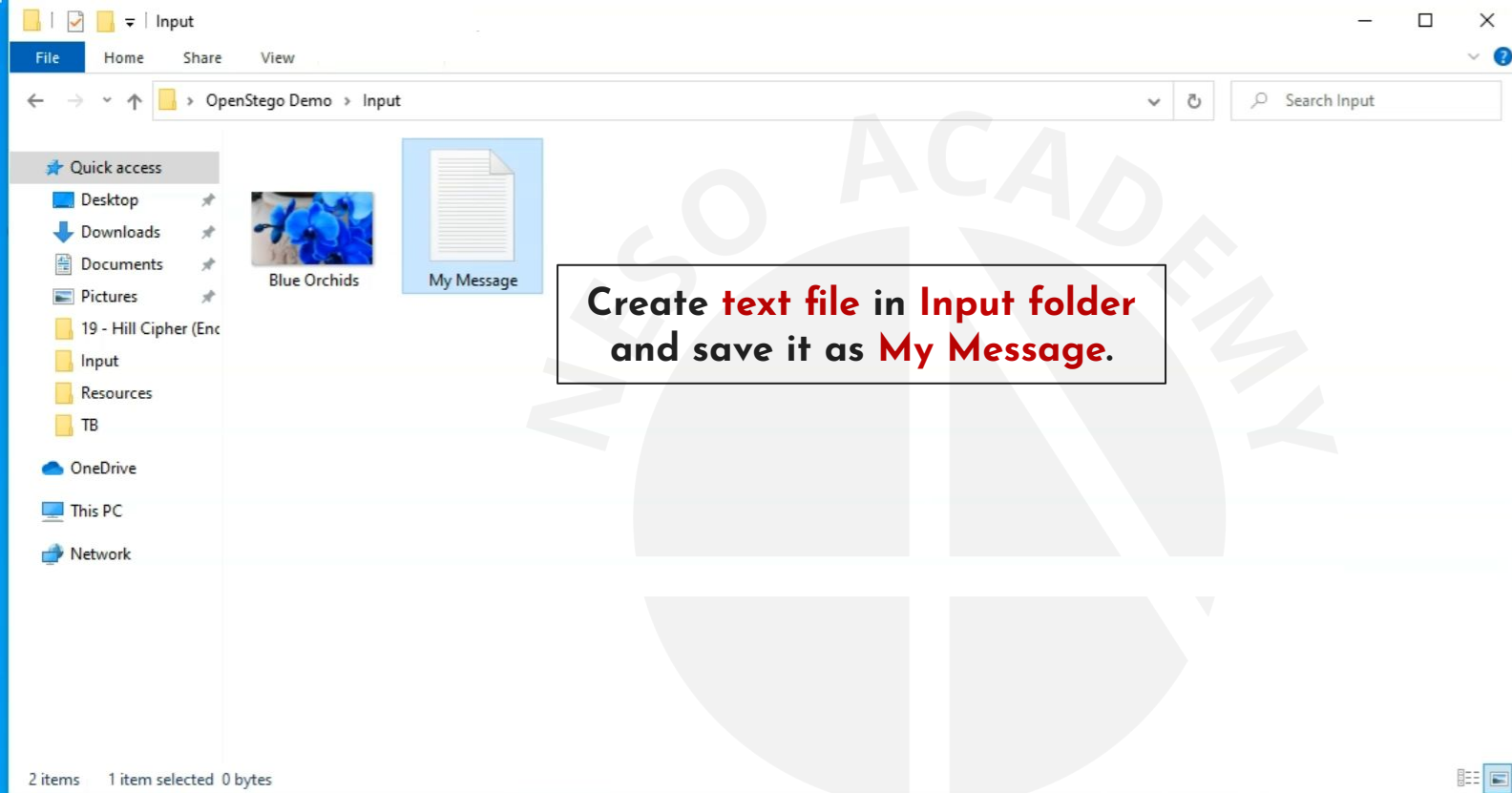
At the bottom of the window, there is a status bar that reads 'Digital Watermarking (Beta)'.

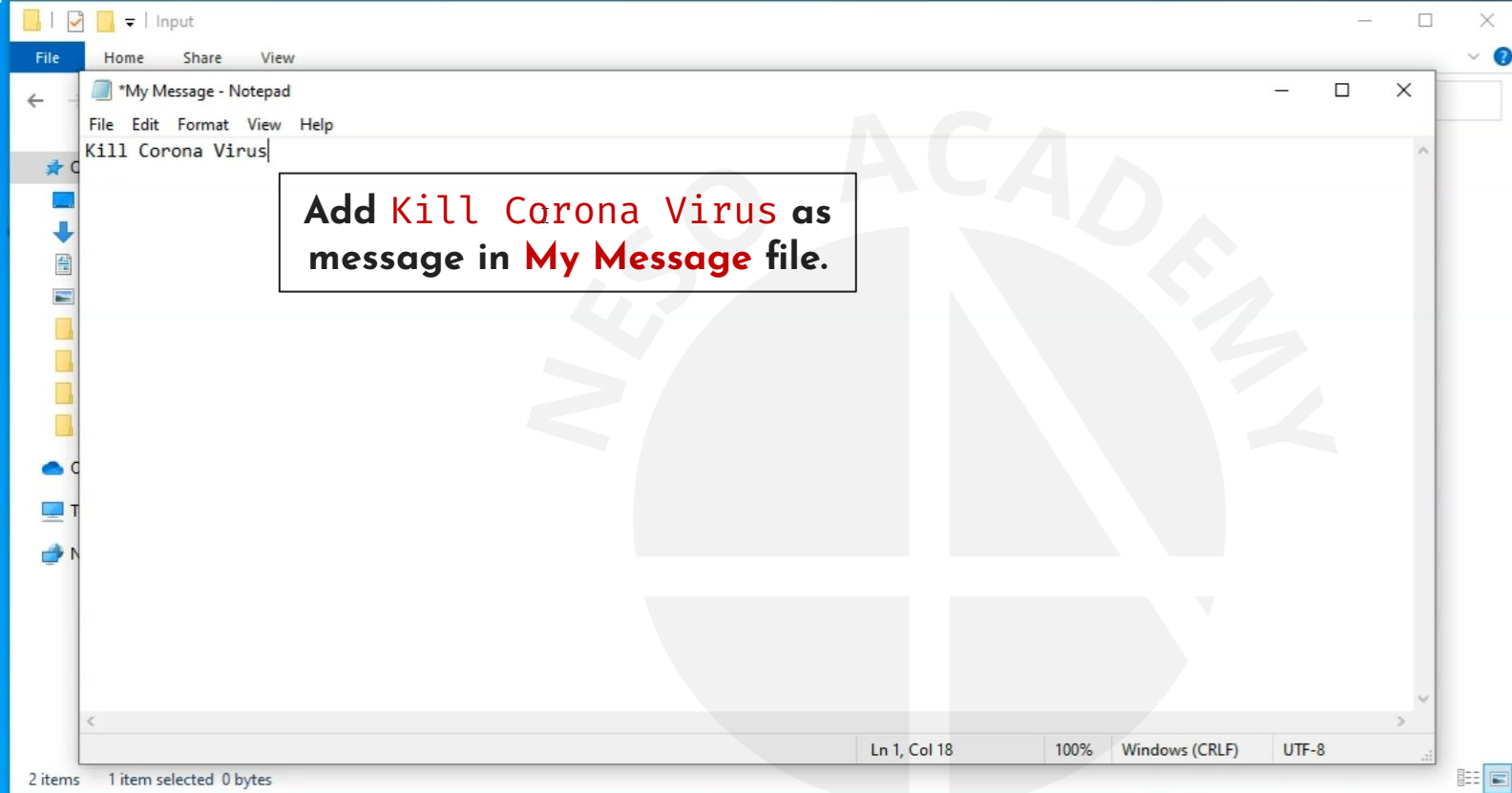


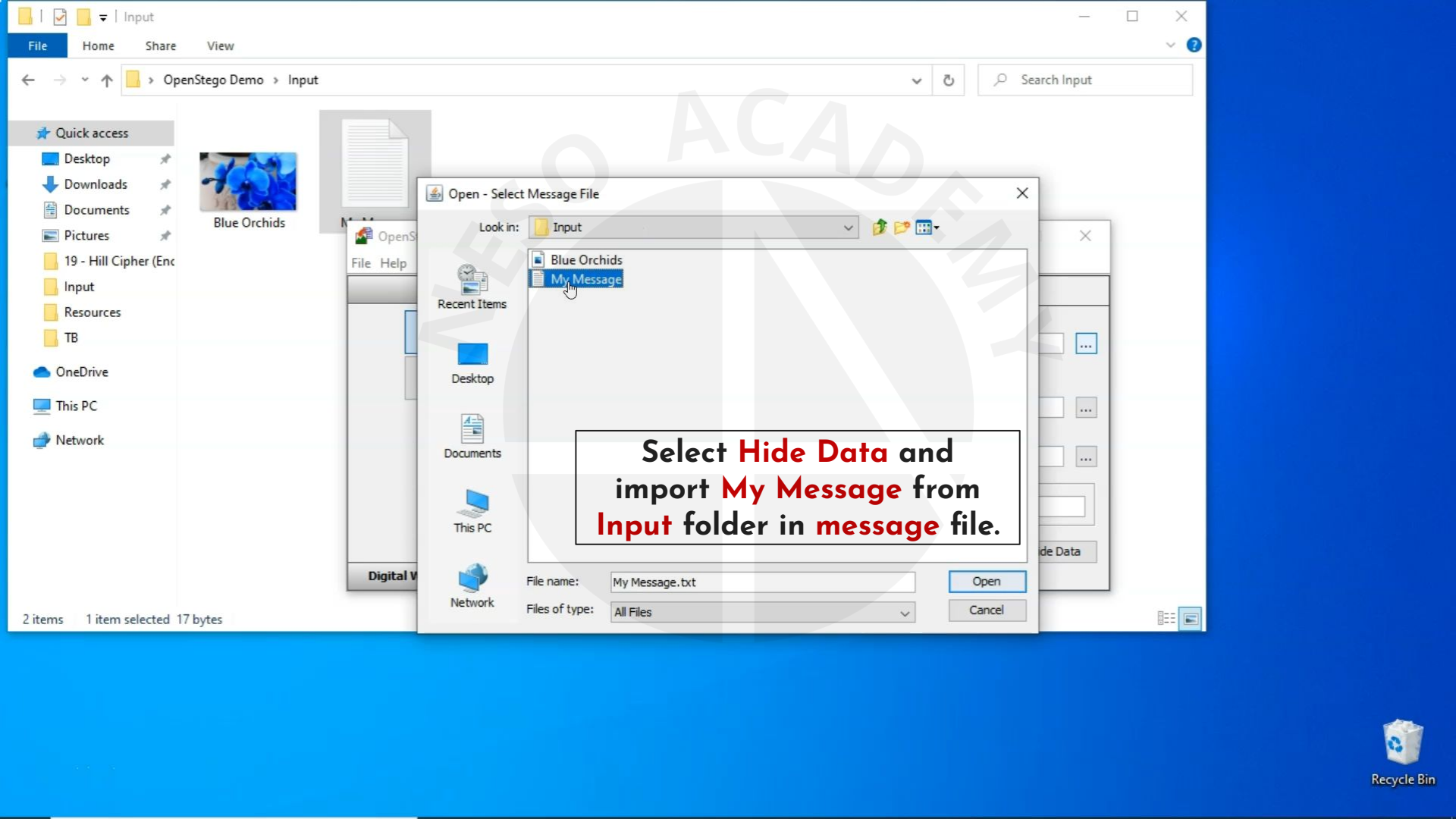
Recycle Bin

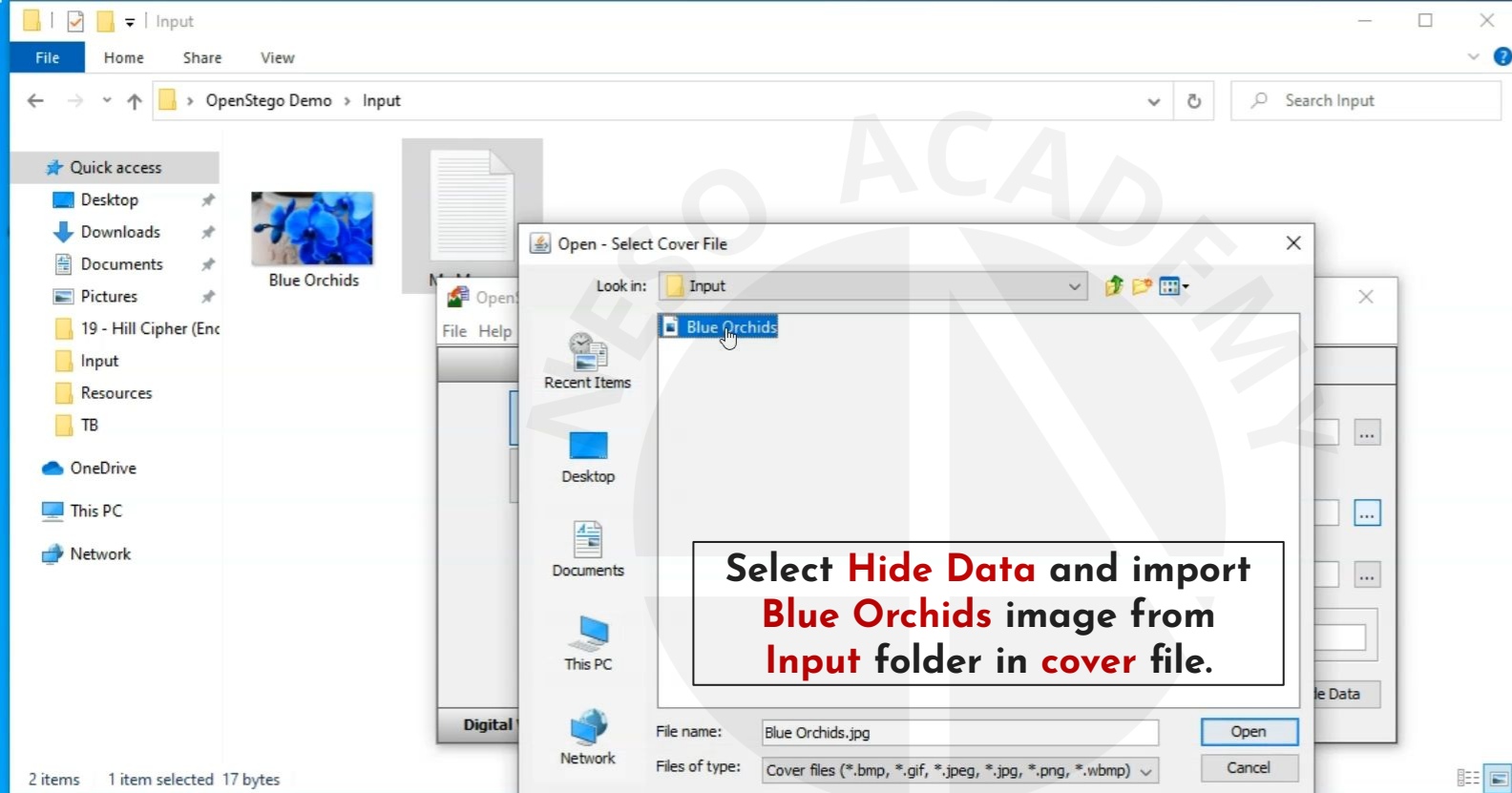


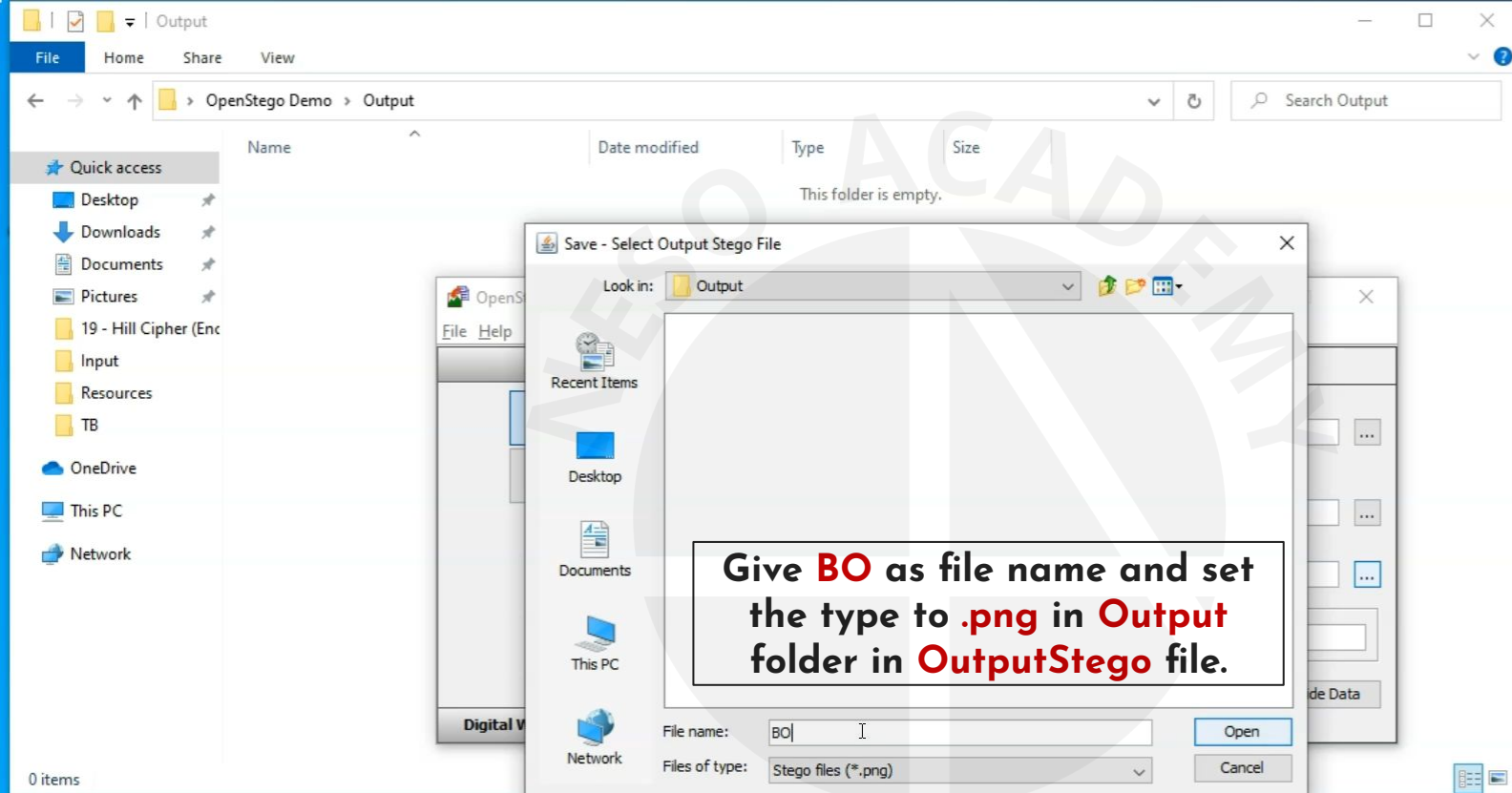
Create **Input** and **Output** folders in **OpenStego Demo** folder. Afterwards, save an **image file** in the **Input** folder. This image file will be used to store the **hidden information**.



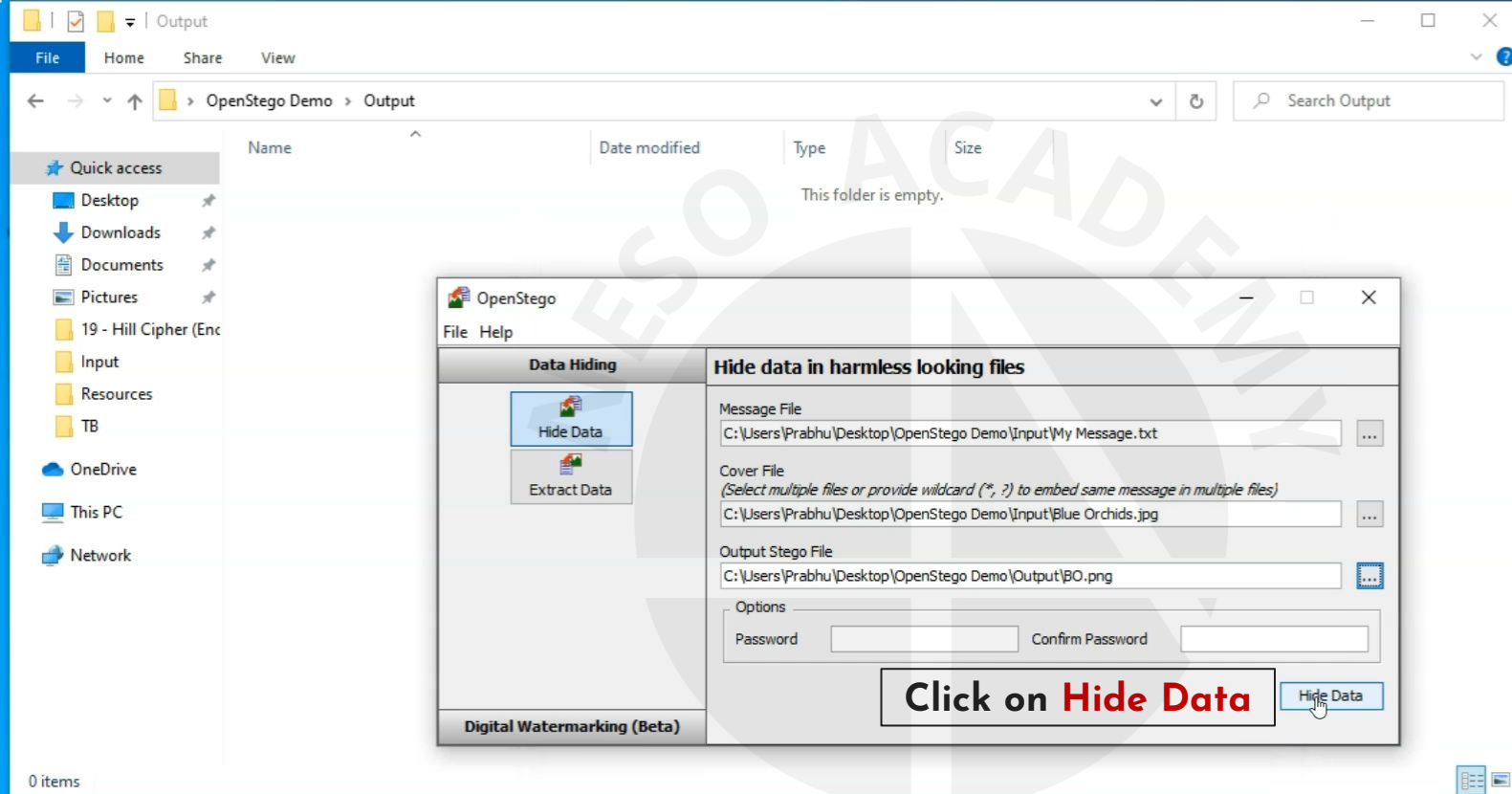


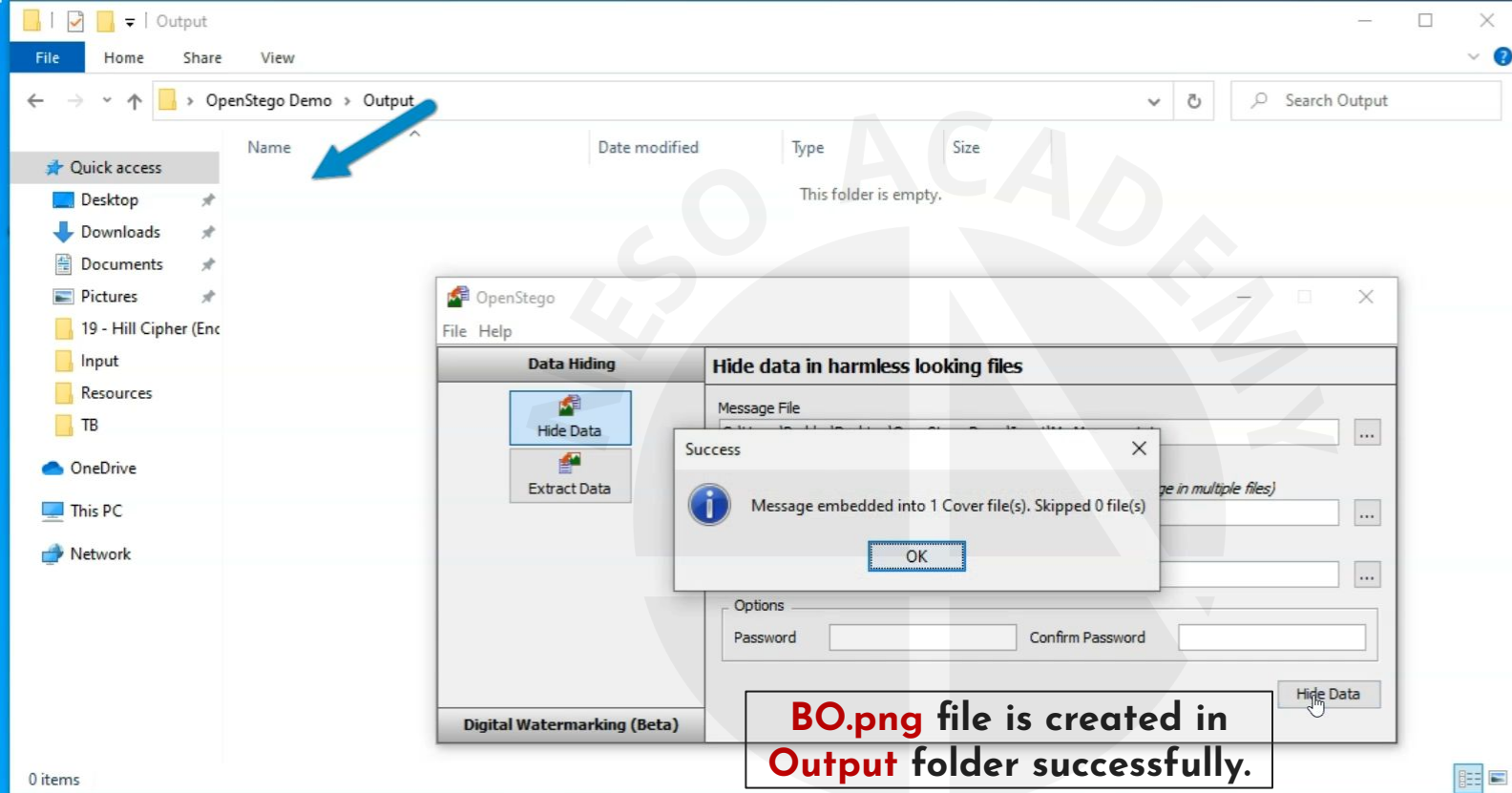


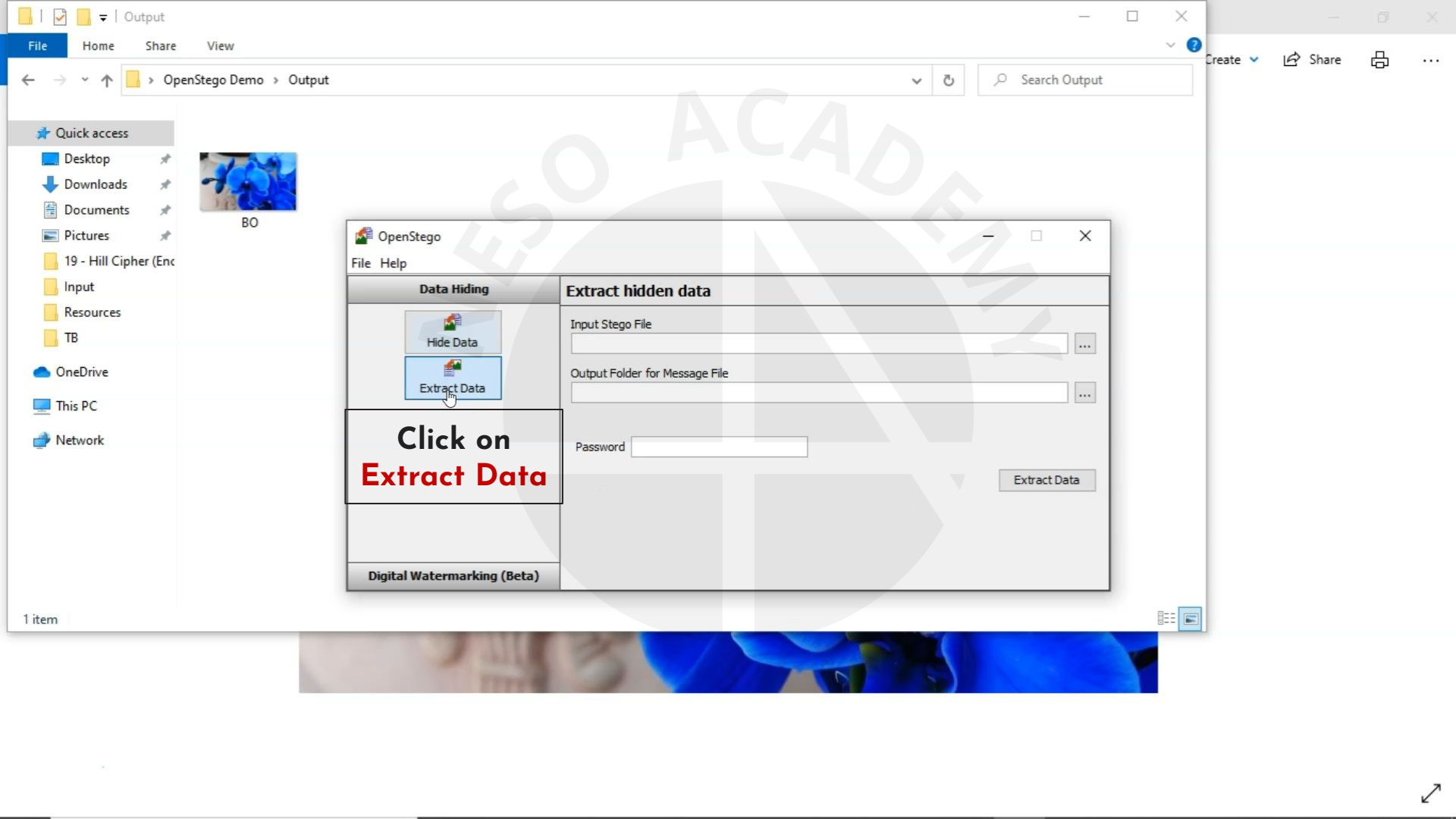


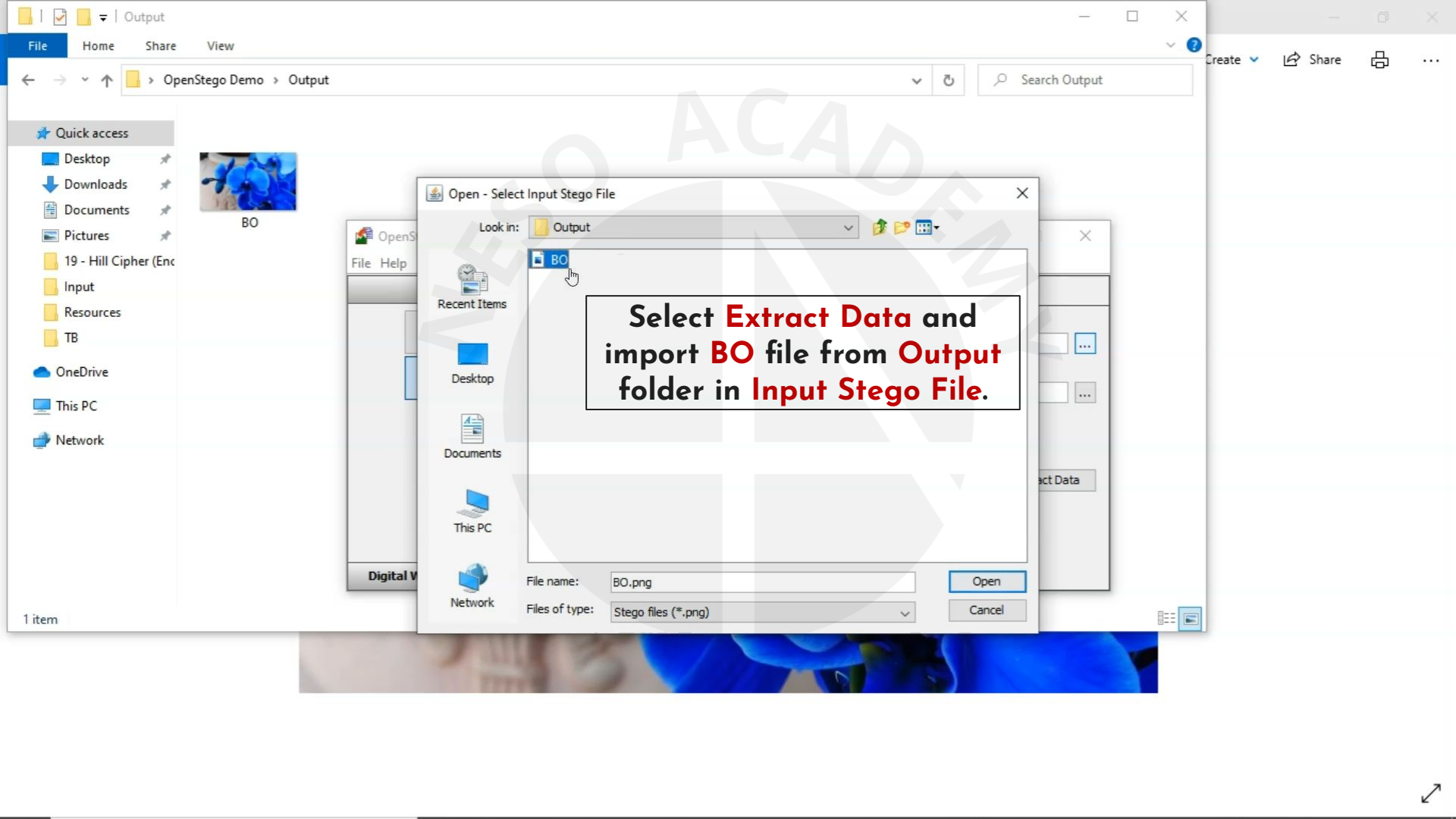


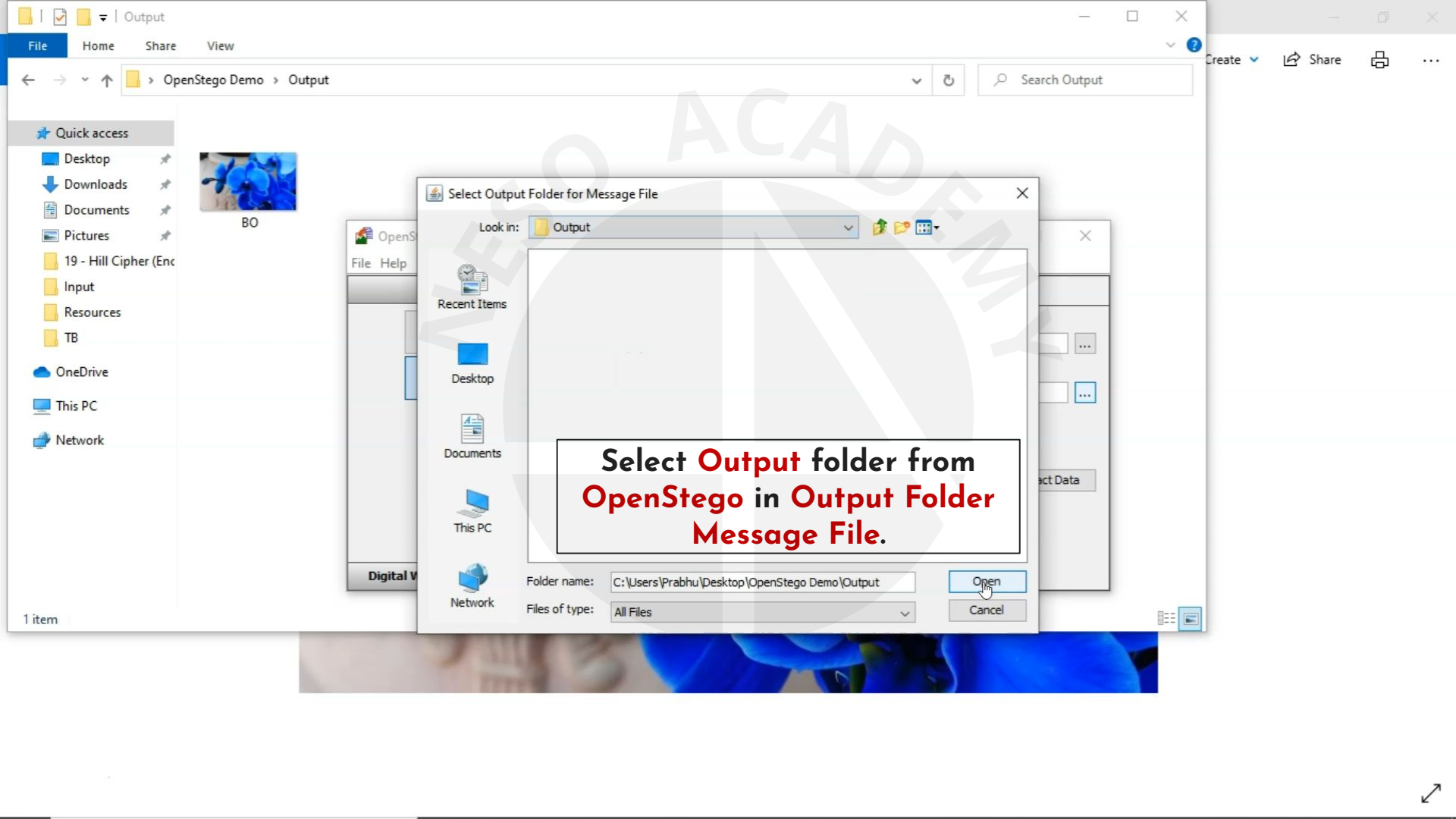
Recycle Bin

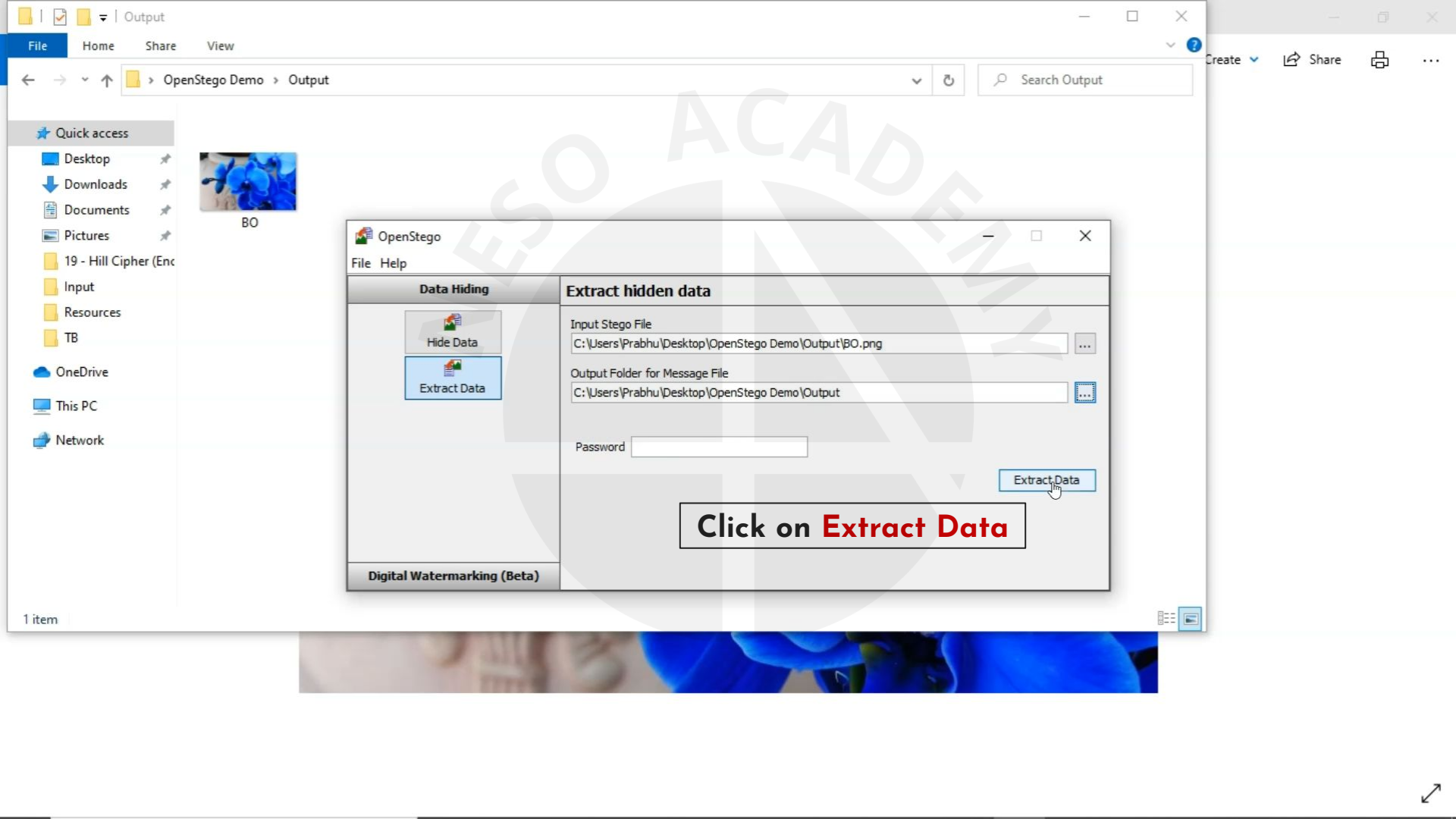












Quick access

- Desktop
- Downloads
- Documents
- Pictures
- 19 - Hill Cipher (Enc
- Input
- Resources
- TB
- OneDrive
- This PC
- Network



BO

OpenStego

File Help

Data Hiding

Hide Data

Extract Data

Extract hidden data

Input Stego File

C:\Users\Prabhu\Desktop\OpenStego Demo\Output\BO.png

Output Folder for Message File

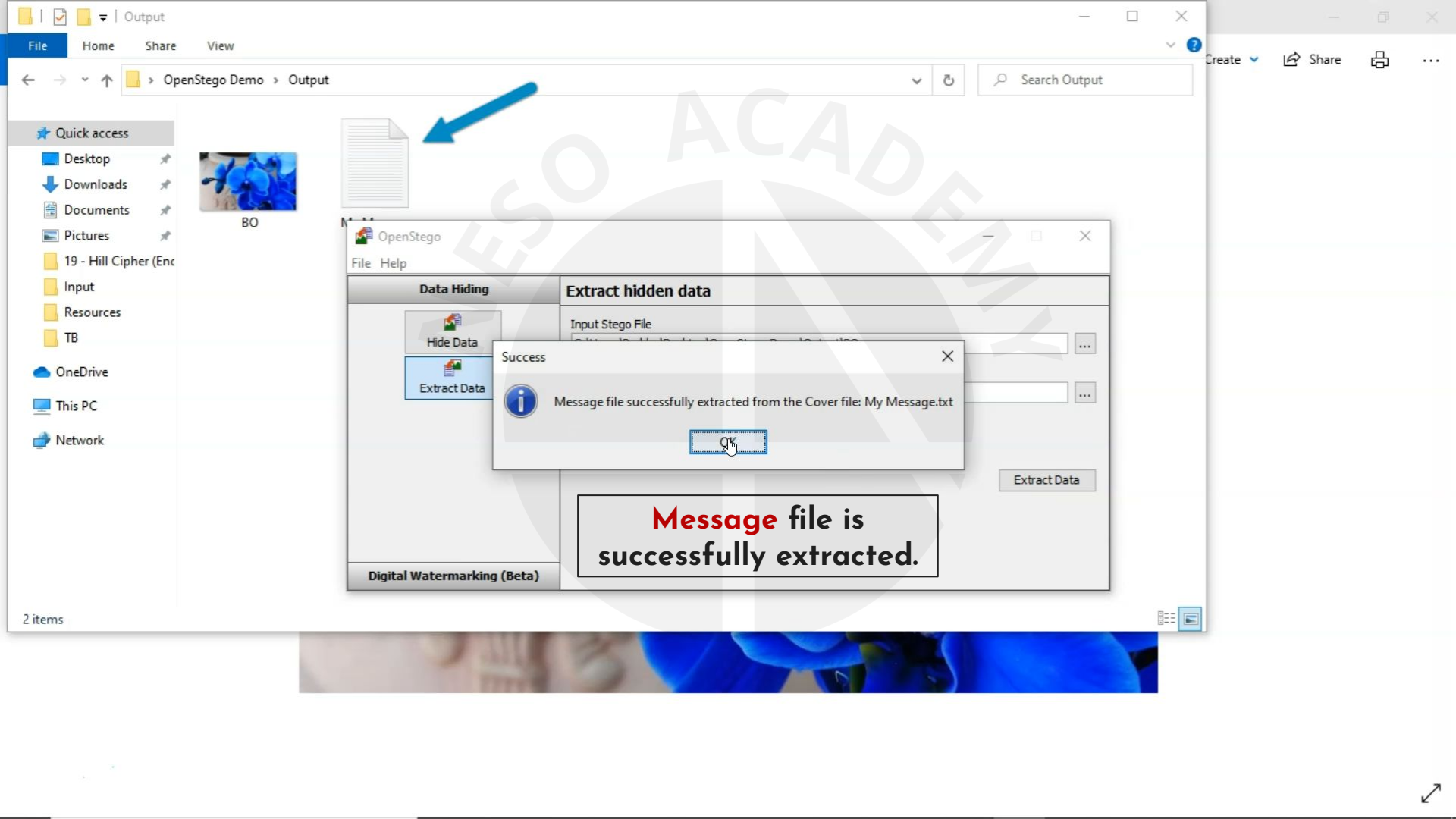
C:\Users\Prabhu\Desktop\OpenStego Demo\Output

Password

Extract Data

Digital Watermarking (Beta)

Click on **Extract Data**



Quick access

- Desktop
- Downloads
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- 19 - Hill Cipher (Enc
- Input
- Resources
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- Network



BO



OpenStego

File Help

Data Hiding

Extract hidden data

Input Stego File

Hide Data

Extract Data

Success

Message file successfully extracted from the Cover file: My Message.txt

Extract Data

Digital Watermarking (Beta)

Message file is
successfully extracted.



Cryptography and Network Security

Cryptography (Solved Questions)

Question 1

In which type of cryptography, sender and receiver uses same key for encryption and decryption.

- (a) Public Key Cryptography
- (b) Private Key Cryptography
- (c) Symmetric Cryptography
- (d) Asymmetric Cryptography

Question 1

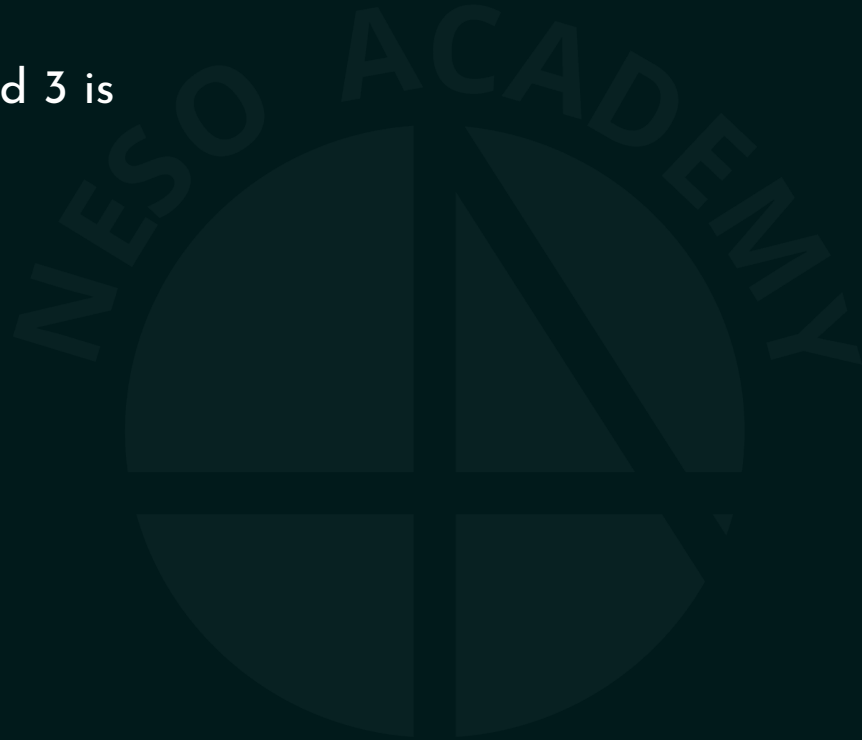
In which type of cryptography, sender and receiver uses same key for encryption and decryption.

- (a) Public Key Cryptography
- (b) Private Key Cryptography
- (c) Symmetric Cryptography
- (d) Asymmetric Cryptography

Question 2

The output of $19 \bmod 3$ is

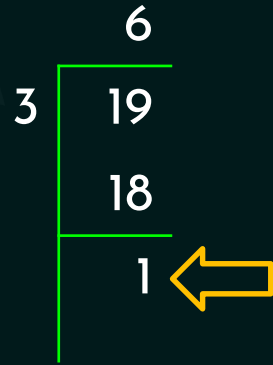
- (a) 19
- (b) 13
- (c) 3
- (d) 1



Question 2

The output of $19 \bmod 3$ is

- (a) 19
- (b) 13
- (c) 3
- (d) 1



A long division diagram showing 19 divided by 3. The divisor 3 is on the left, and the dividend 19 is on the right. A horizontal line is drawn under the 19. The quotient 6 is written above the line. Below the line, the product 18 is written, and the remainder 1 is written below that. A yellow arrow points to the remainder 1.

$$\begin{array}{r} 6 \\ 3 \overline{) 19} \\ \underline{18} \\ 1 \end{array}$$

Question 3

An attacker sits between the sender and receiver and captures the information and retransmits to the receiver after some time without altering the information. This attack is called as _____.

[UGC NET CS 2016]

- (a) Denial of Service attack
- (b) Masquerade attack
- (c) Simple attack
- (d) Complex attack

Question 3

An attacker sits between the sender and receiver and captures the information and retransmits to the receiver after some time without altering the information. This attack is called as _____.

[UGC NET CS 2016]

- (a) Denial of Service attack
- (b) Masquerade attack
- (c) Simple attack
- (d) Complex attack

Question 4

In a columnar transposition cipher, the plain text is "the tomato is a plant in the night shade family", keyword is "TOMATO". The cipher text is

[ISRO CS 2020]

- (a) "TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX"
- (b) "TINESAX / EOAHTFX / MAIIAIX / HTLTHEY / TAPNGDL / OSTNHMX"
- (c) "TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / OSTNHMX / TAPNGDL"
- (d) "EOAHTFX / TINESAX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX"

Solution

t o m a t o
4 3 2 1 4 3

t	h	e	t	o	m
a	t	o	i	s	a
p	l	a	n	t	i
n	t	h	e	n	i
g	h	t	s	h	a
d	e	f	a	m	i
l	y	x	x	x	x

Solution

t	o	m	a	t	o
4	3	2	1	4	3
t	h	e	t	o	m
a	t	o	i	s	a
p	l	a	n	t	i
n	t	h	e	n	i
g	h	t	s	h	a
d	e	f	a	m	i
l	y	x	x	x	x

Solution

t	o	m	a	t	o
4	3	2	1	4	3
t	h	e	t	o	m
a	t	o	i	s	a
p	l	a	n	t	i
n	t	h	e	n	i
g	h	t	s	h	a
d	e	f	a	m	i
l	y	x	x	x	x

Solution

t	o	m	a	t	o
4	3	2	1	4	3
t	h	e	t	o	m
a	t	o	i	s	a
p	l	a	n	t	i
n	t	h	e	n	i
g	h	t	s	h	a
d	e	f	a	m	i
l	y	x	x	x	x

Solution

t	o	m	a	t	o
4	3	2	1	4	3
t	h	e	t	o	m
a	t	o	i	s	a
p	l	a	n	t	i
n	t	h	e	n	i
g	h	t	s	h	a
d	e	f	a	m	i
l	y	x	x	x	x

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[ISRO CS 2020]

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- (c) "TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / OSTNHMX / TAPNGDL"
- (d) "EOAHTFX / TINESAX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX"

Question 5

Suppose that everyone in a group of N people wants to communicate secretly with the $N - 1$ others using symmetric key cryptographic system. The communication between any two persons should not be decodable by the others in the group. The number of keys required in the system as a whole to satisfy the confidentiality requirement is

[GATE CS 2015]

- (a) $2N$
- (b) $N(N-1)$
- (c) $\frac{N(N-1)}{2}$
- (d) $(N-1)^2$

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[GATE CS 2015]

(a) $2N$

(b) $N(N-1)$

(c) $\frac{N(N-1)}{2}$

(d) $(N-1)^2$

No of people	Keys
1	0
2	1
3	3
4	6
5	10



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